

An analysis of the Virgo arm cavities imperfections using Machine Learning

Let's break the title down:

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- 1) Cavities imperfections

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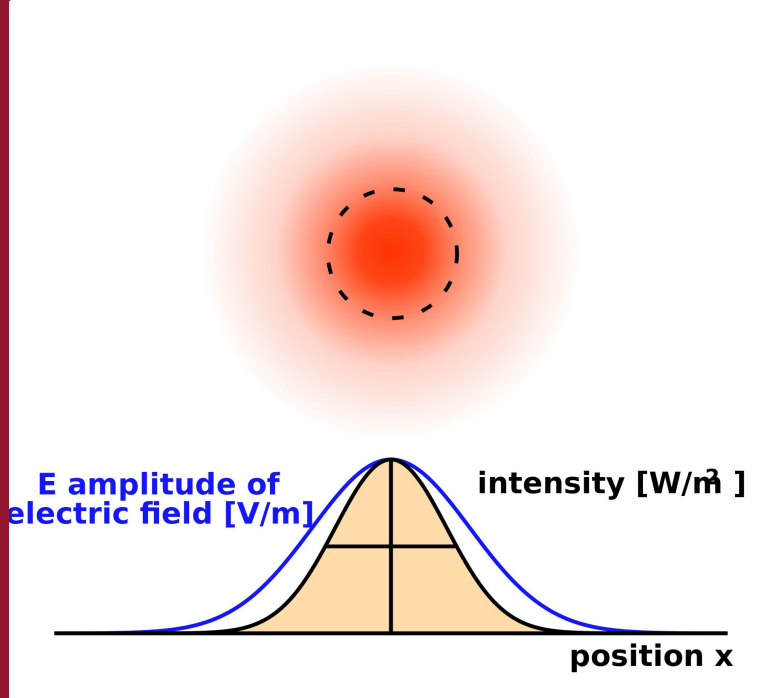
- 1) Cavities imperfections
- 2) What is Machine Learning and why do we need it?

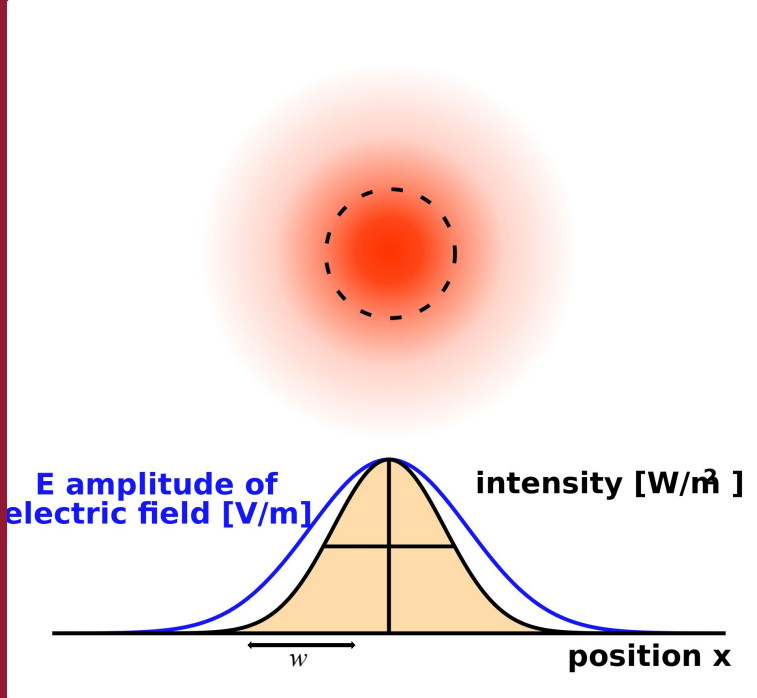
Let's break the title down:

- 1) Cavities imperfections
- 2) What is Machine Learning and why do we need it?
- 3) The analysis

1) Cavities imperfections: mismatching, misalignment and astigmatism

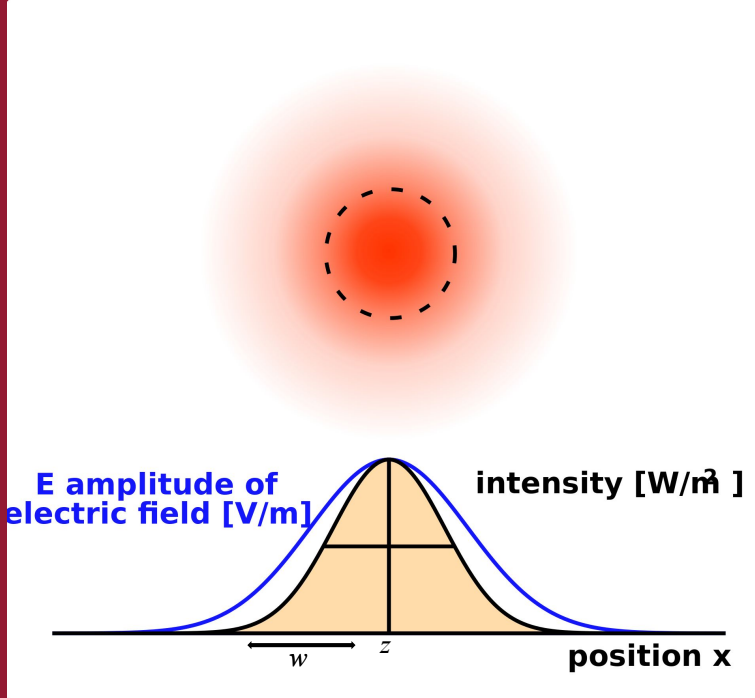
Gaussian beam





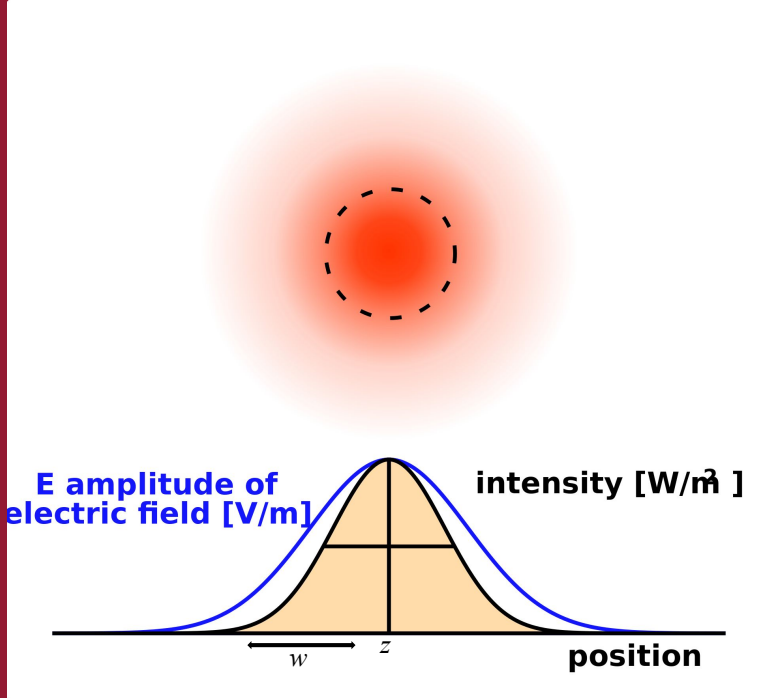
Gaussian beam

- Waist size w



Gaussian beam

- Waist size w
- Waist position z



Gaussian beam

- Waist size w
- Waist position z

→ The cavity has the same parameters (*target*)!

If the parameters match

If the parameters match → the beam is *coupled* to the cavity

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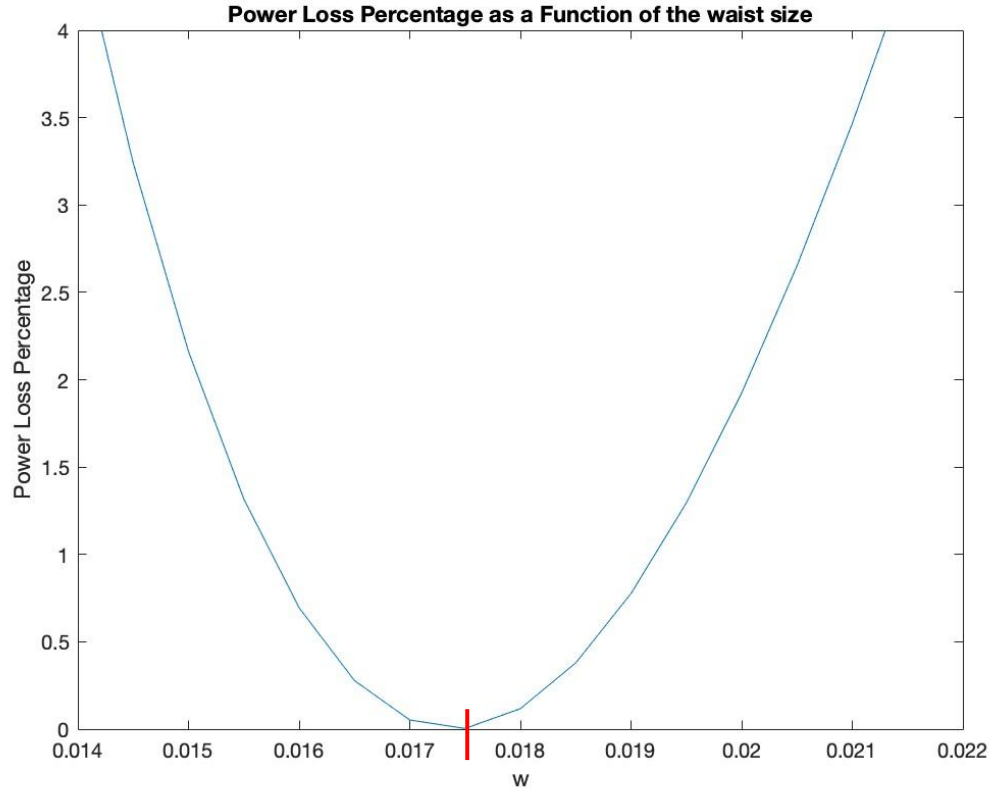
If the parameters **don't** match

If the parameters match → the beam is *coupled* to the cavity

If the parameters **don't** match → *mode mismatch*

If the parameters match → the beam is *coupled* to the cavity

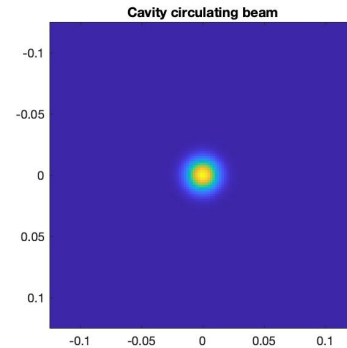
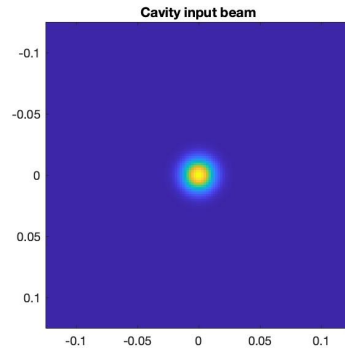
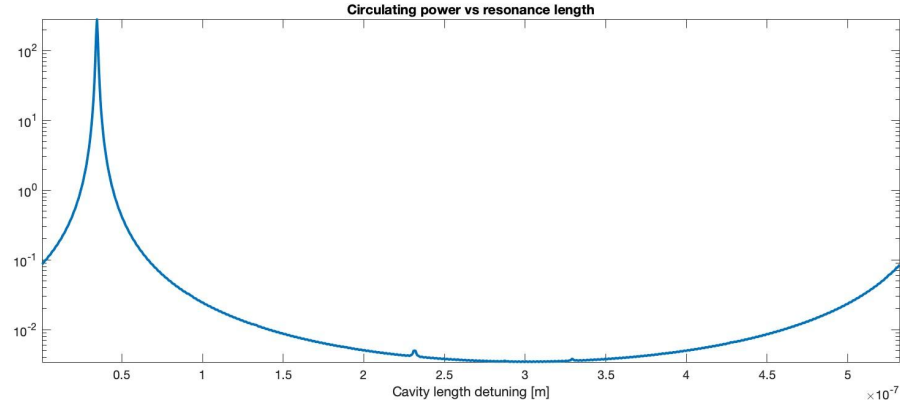
If the parameters **don't** match → *mode mismatch*
→ power loss



Target parameters

Target parameters

—→ *Fundamental mode* of the cavity



Target parameters

→ *Fundamental mode* of the cavity

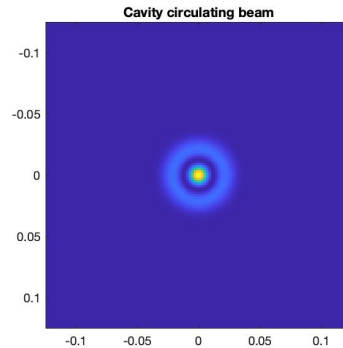
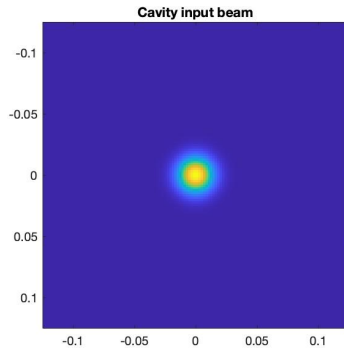
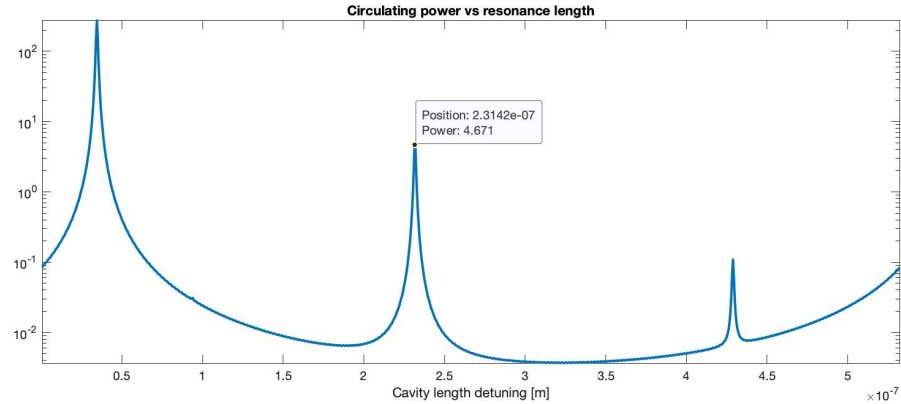
Different parameters

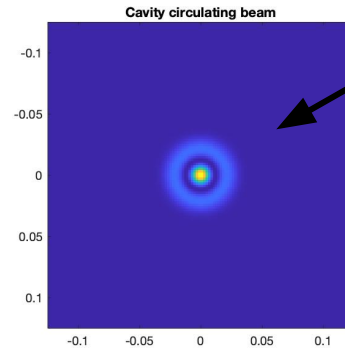
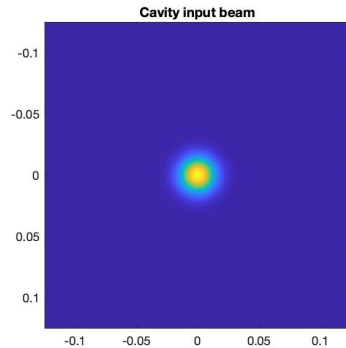
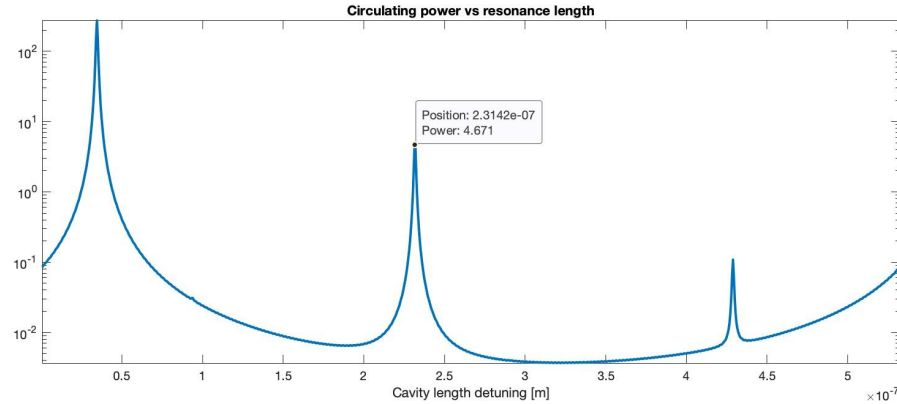
Target parameters

→ *Fundamental mode* of the cavity

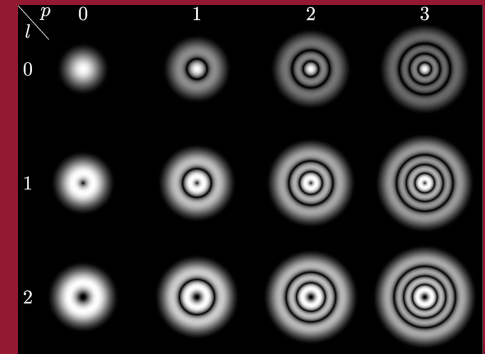
Different parameters

→ *High Order Modes* (HOM)





Laguerre-Gauss mode



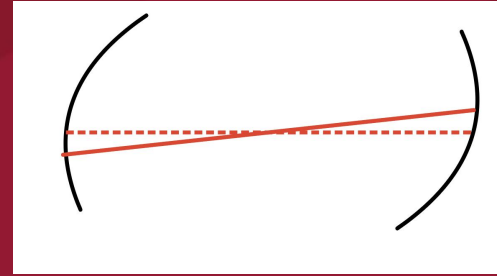
Misalignment: input axis not aligned to cavity axis

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- Mirrors tilted in same direction

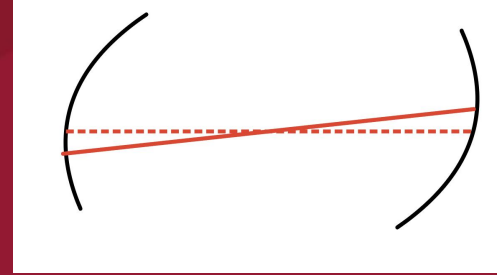
Misalignment: input axis not aligned to cavity axis

- Mirrors tilted in same direction $\longrightarrow$ *tilt*



Misalignment: input axis not aligned to cavity axis

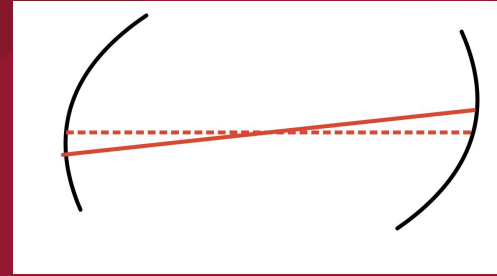
- Mirrors tilted in same direction $\longrightarrow$ *tilt*



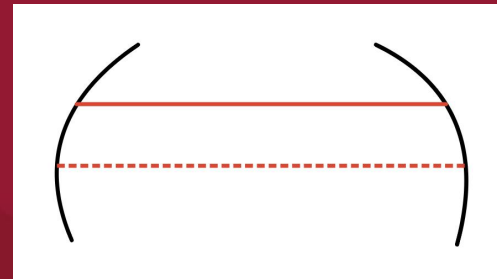
- Mirrors tilted in opposite direction

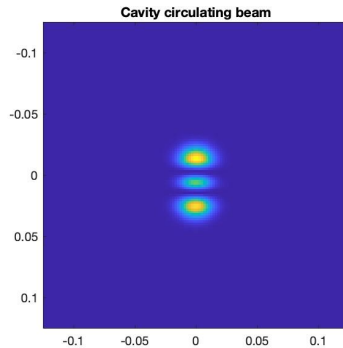
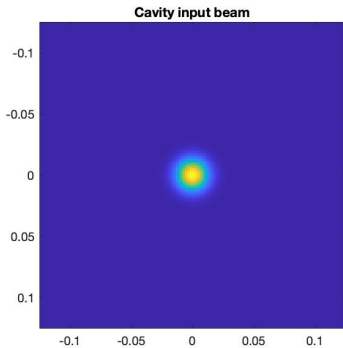
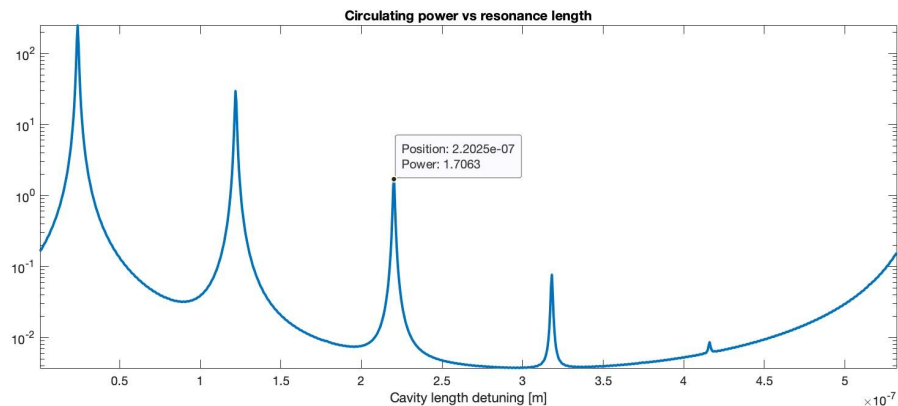
Misalignment: input axis not aligned to cavity axis

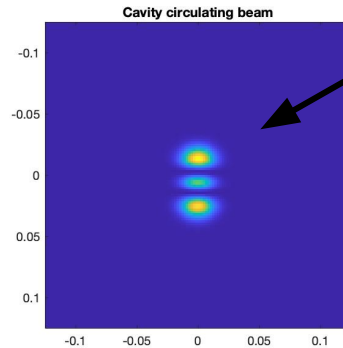
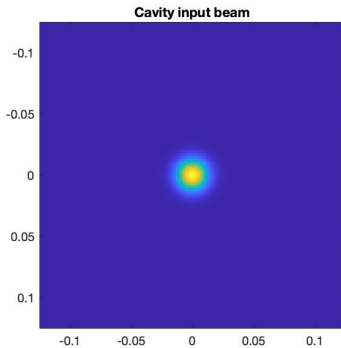
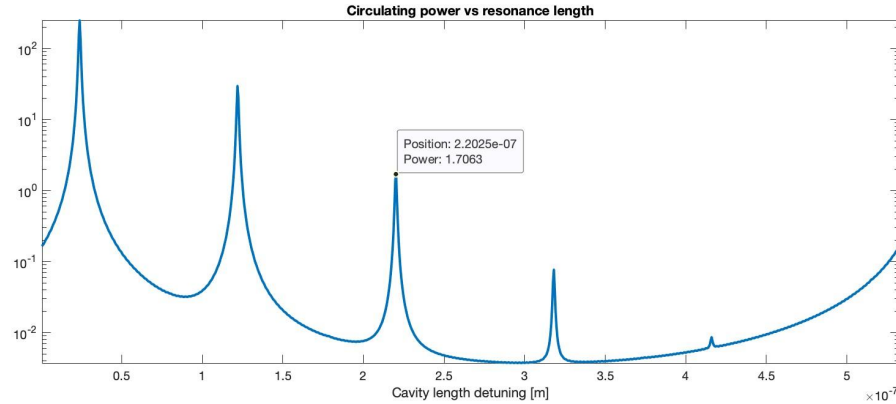
- Mirrors tilted in same direction $\longrightarrow$ *tilt*



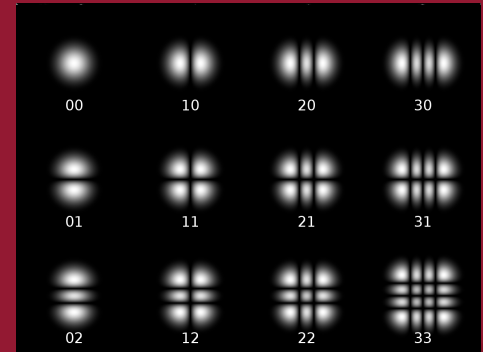
- Mirrors tilted in opposite direction $\longrightarrow$ *shift*

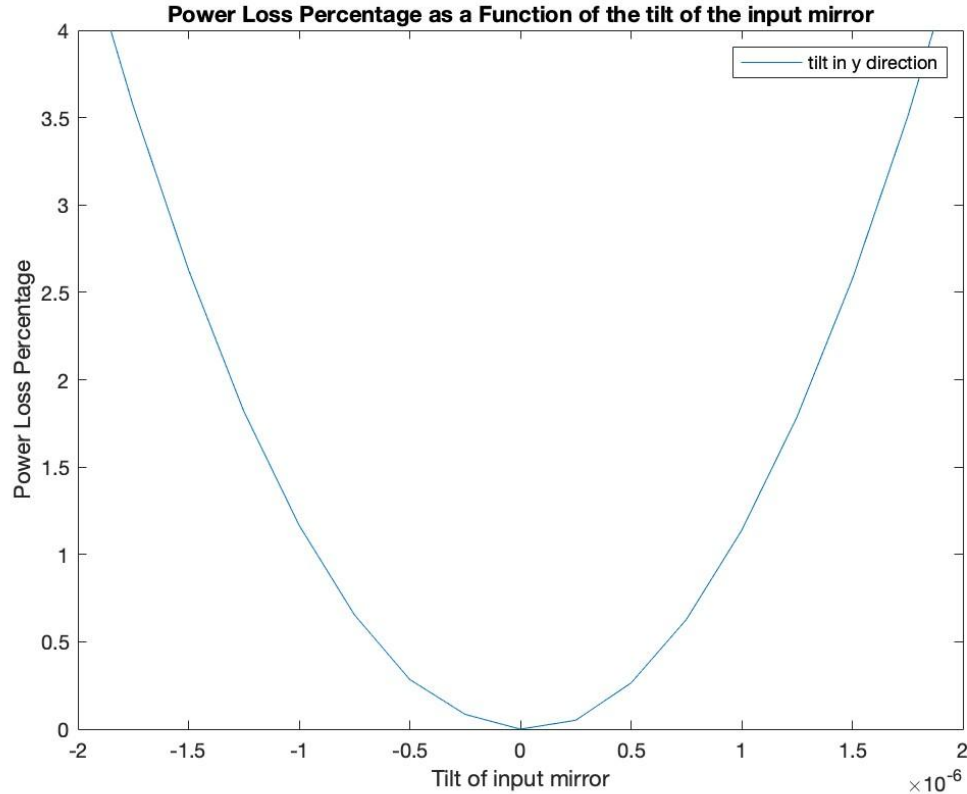






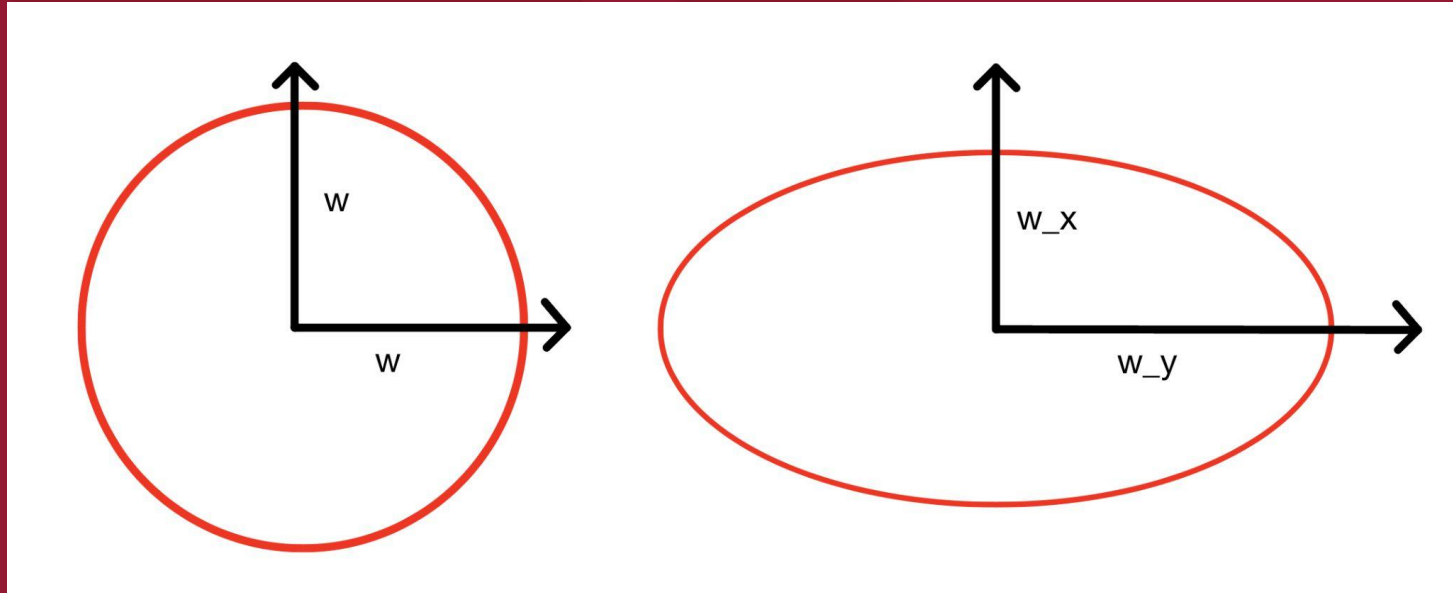
Hermite-Gauss mode

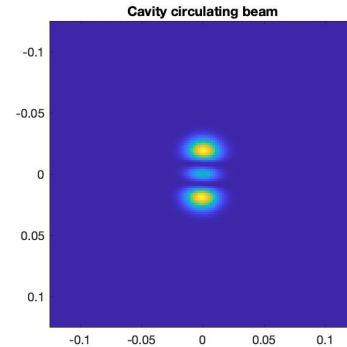
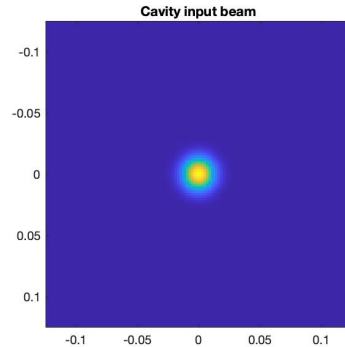
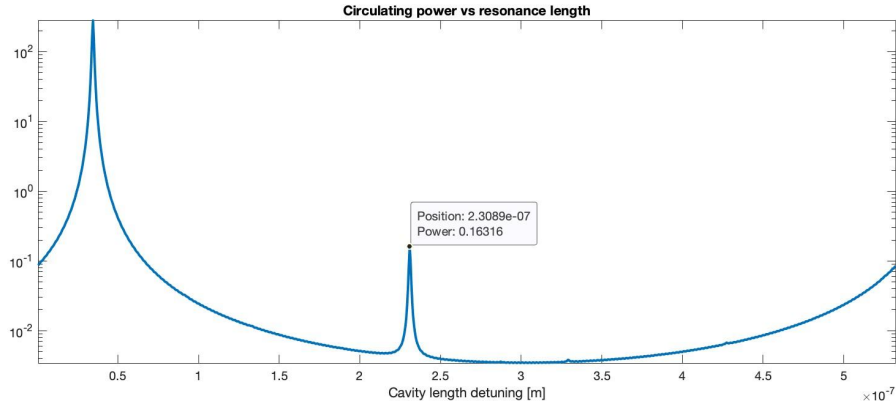


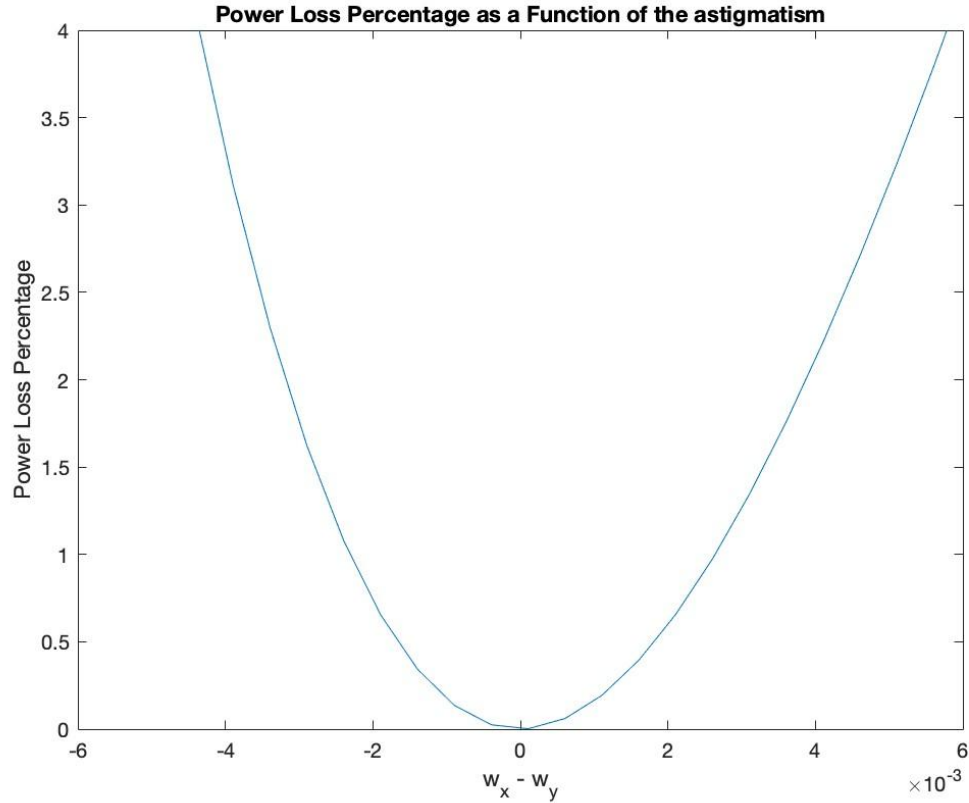


Astigmatism: asymmetric beam

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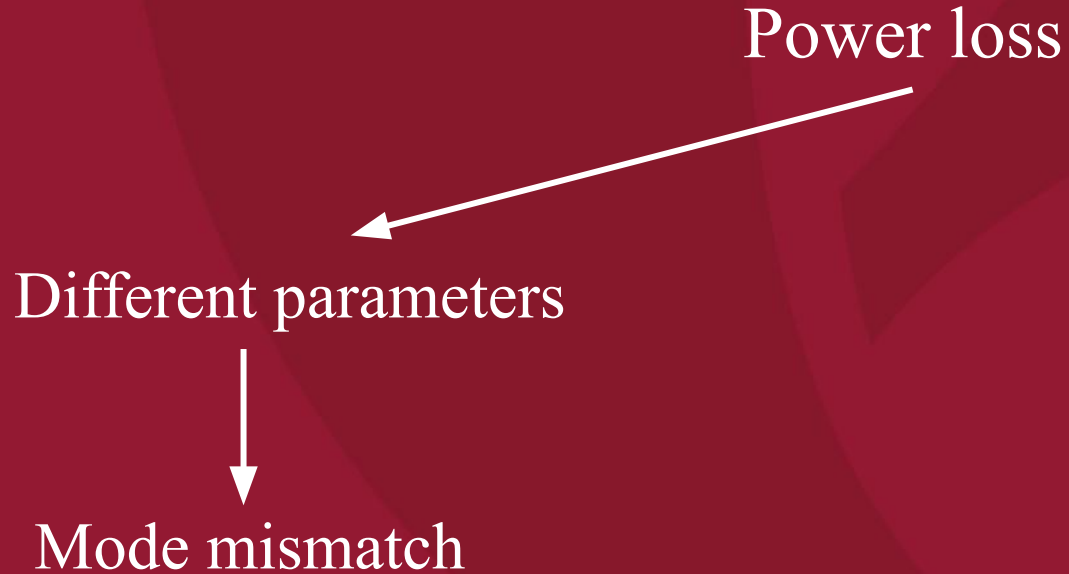


Power loss

Power loss



Different parameters



Power loss



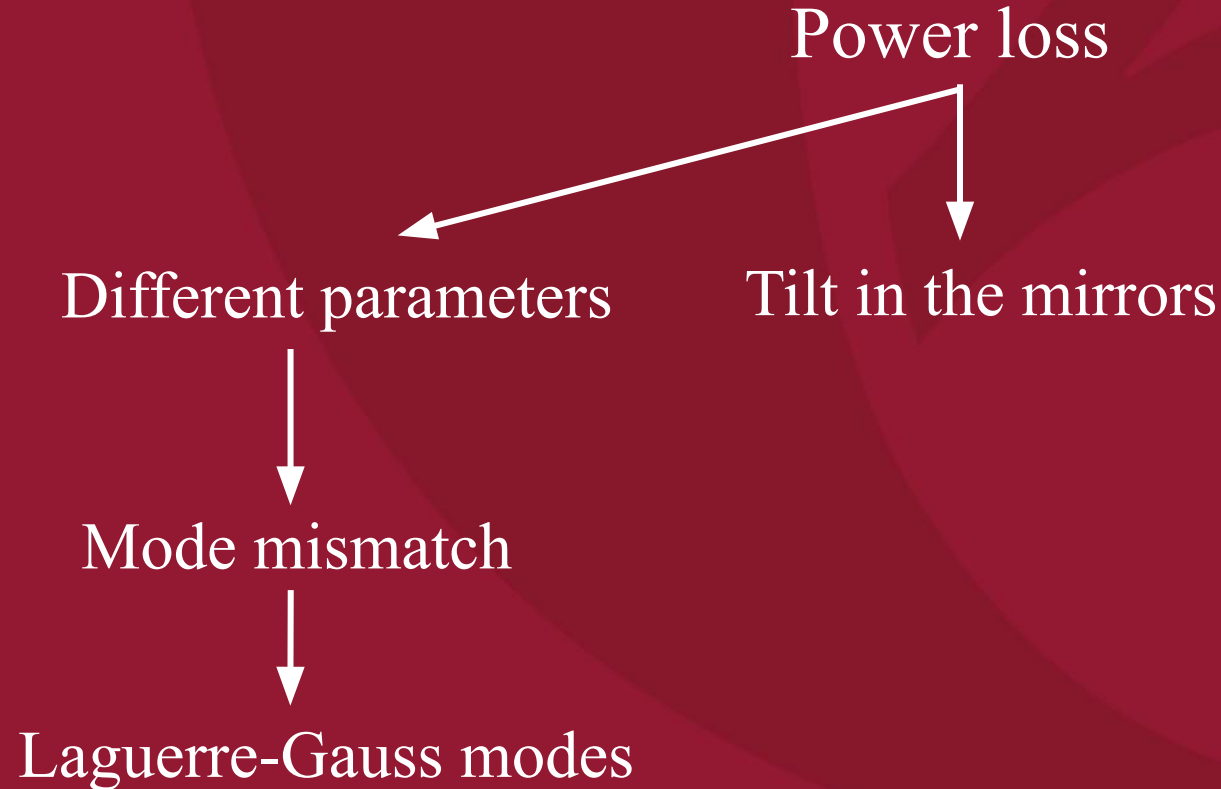
Different parameters

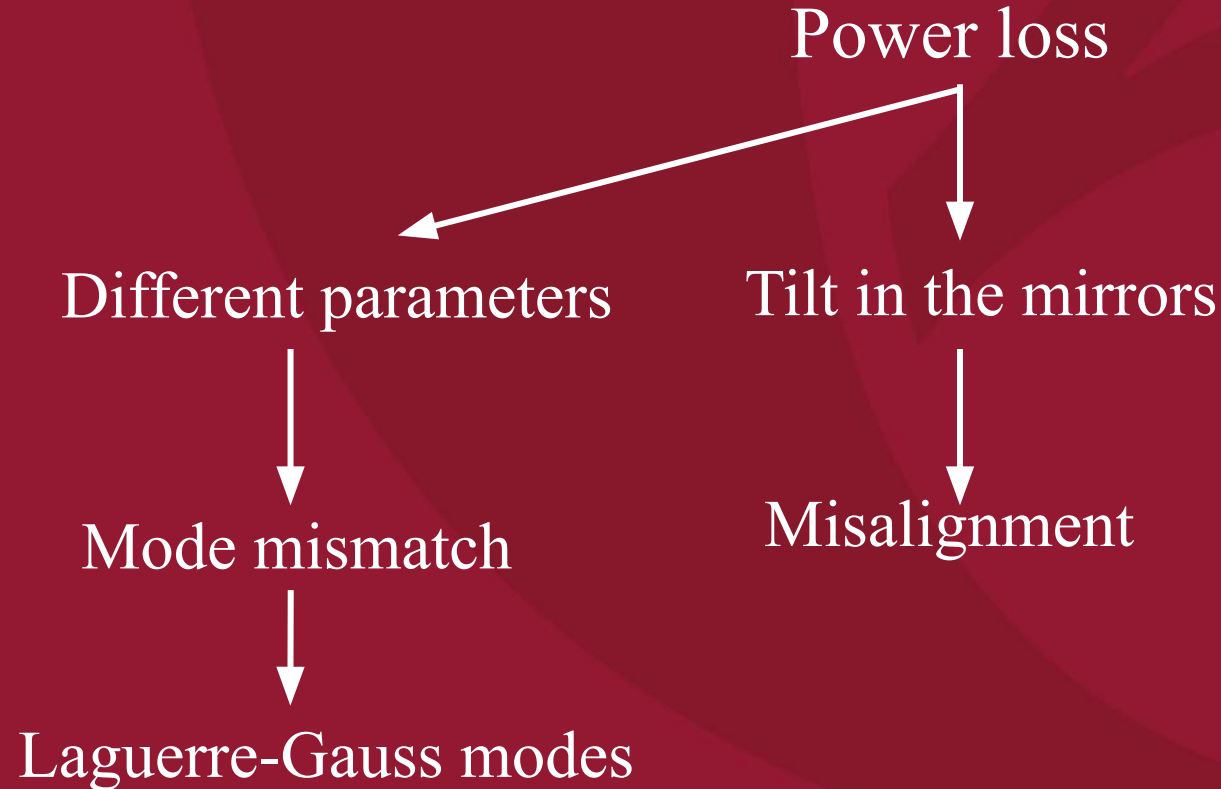


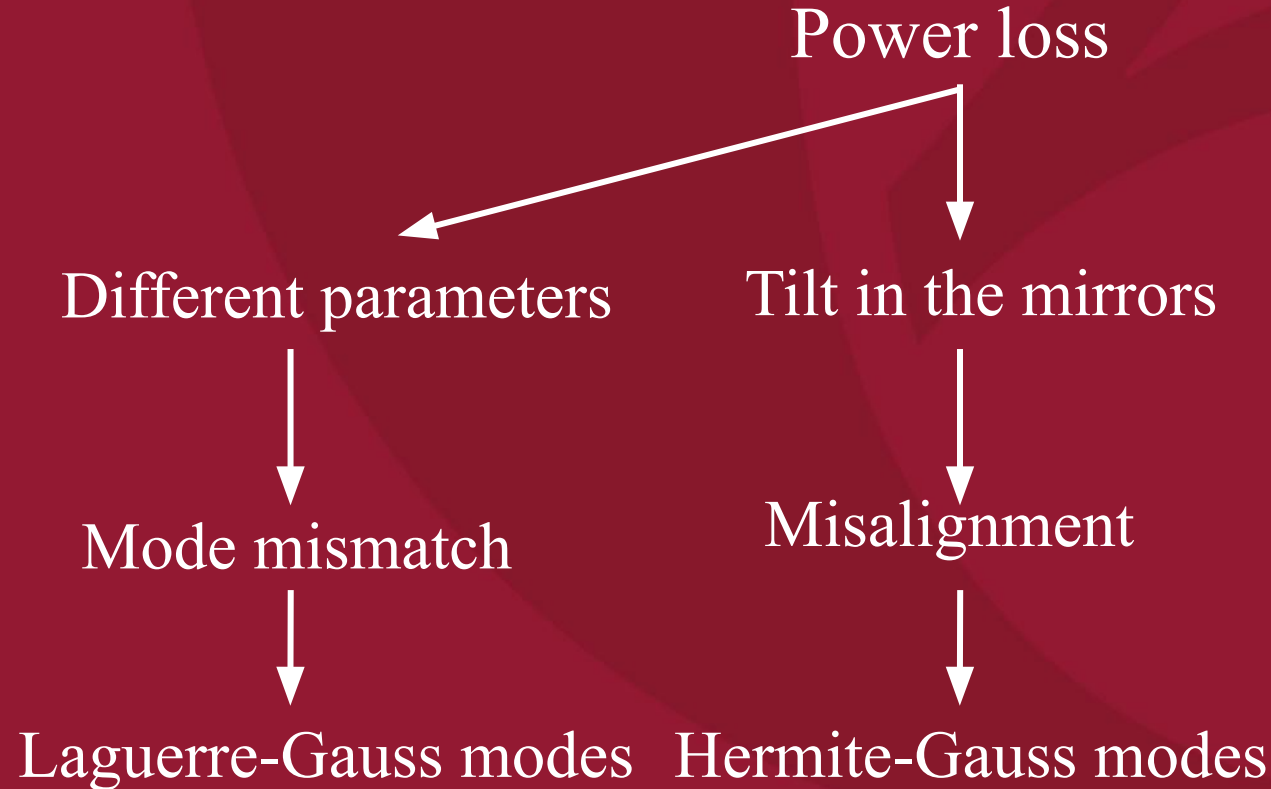
Mode mismatch

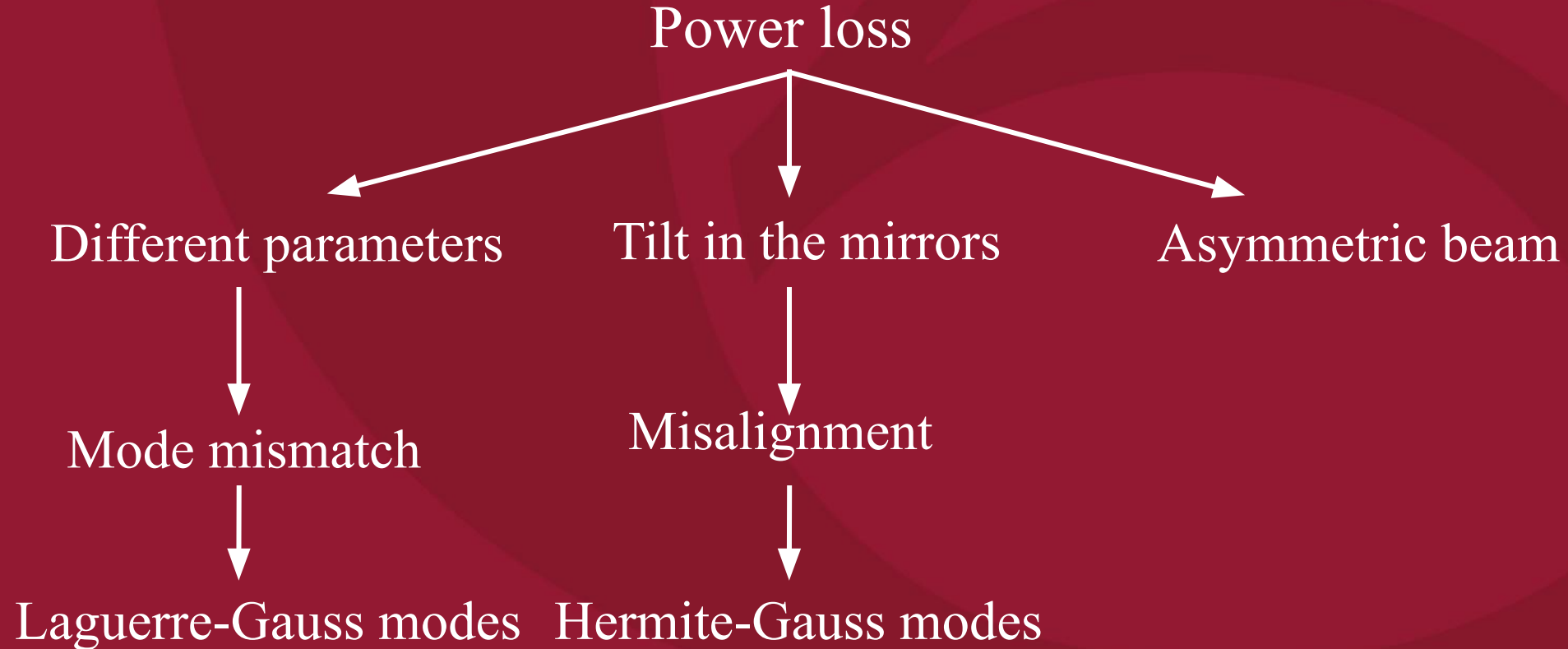


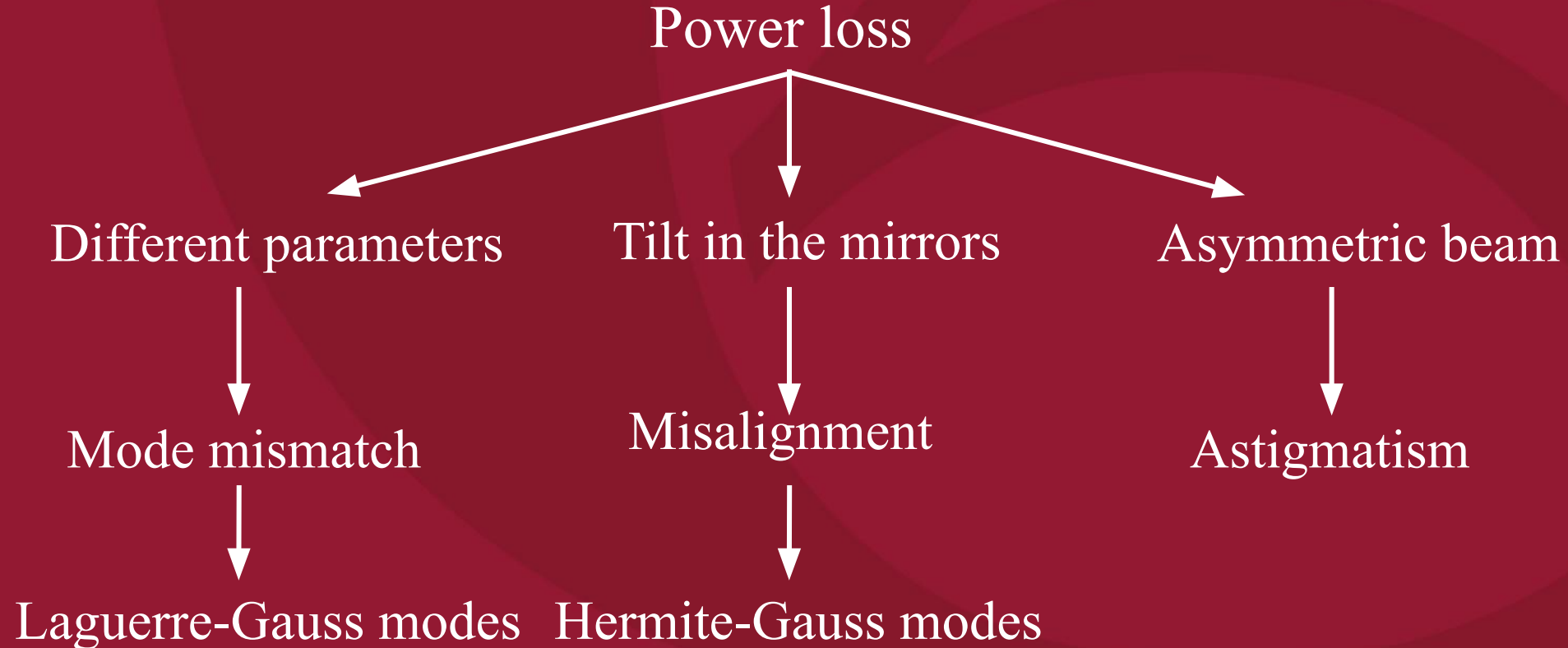
Laguerre-Gauss modes

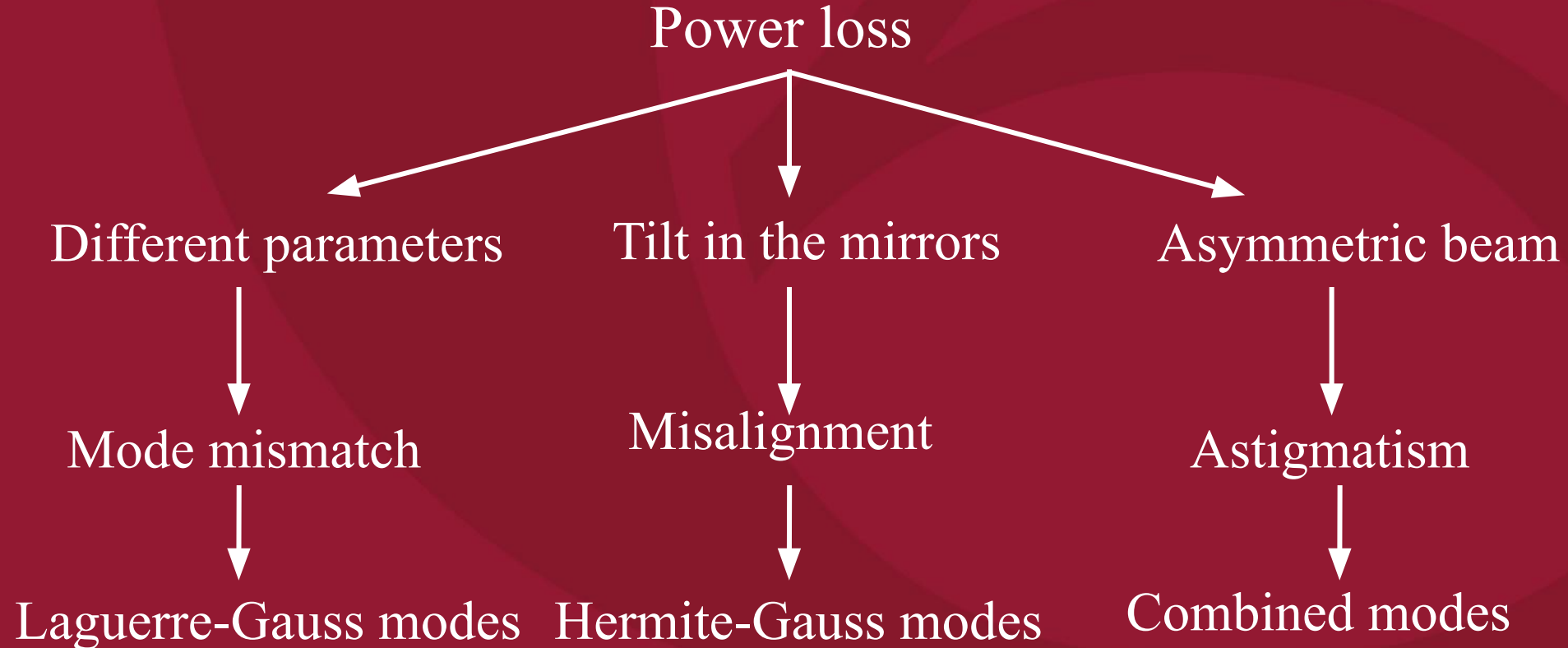


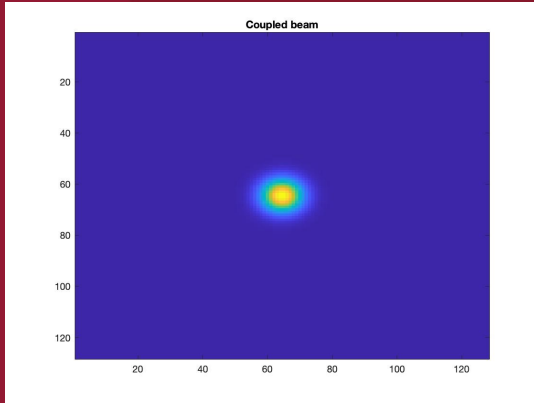


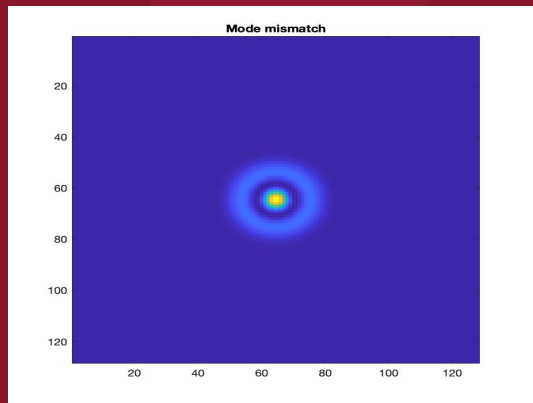
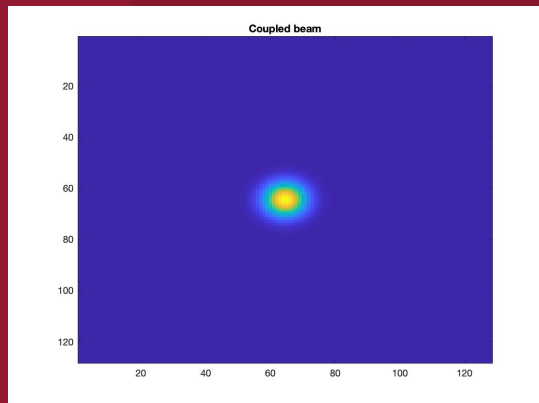




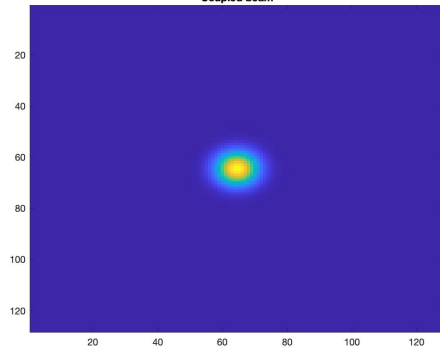




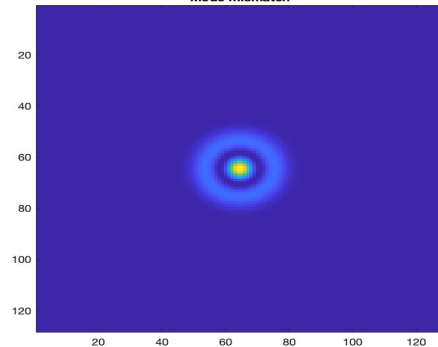




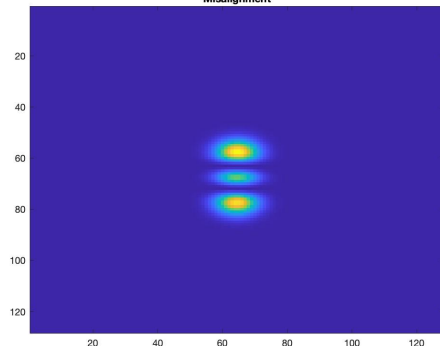
Coupled beam



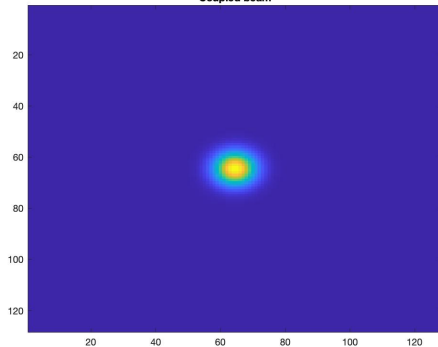
Mode mismatch



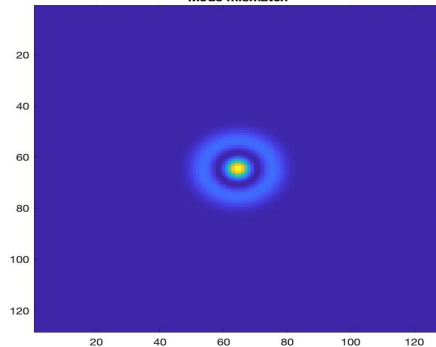
Misalignment



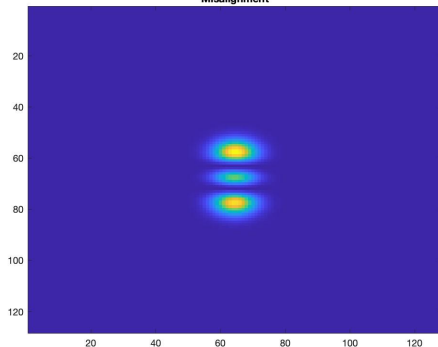
Coupled beam



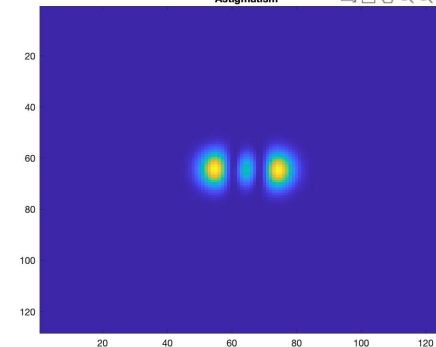
Mode mismatch

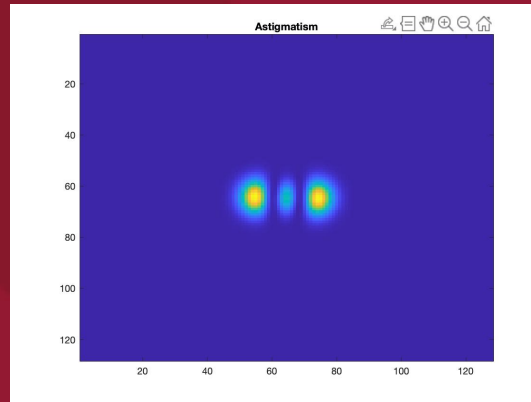
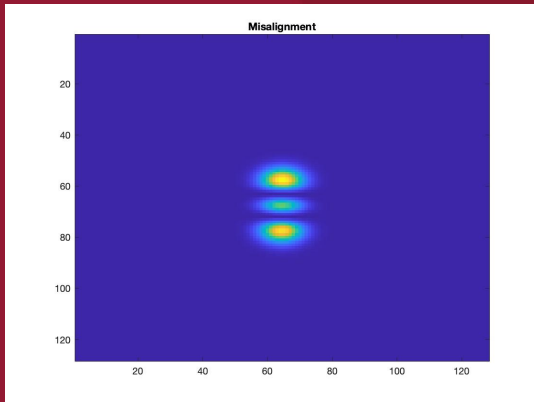
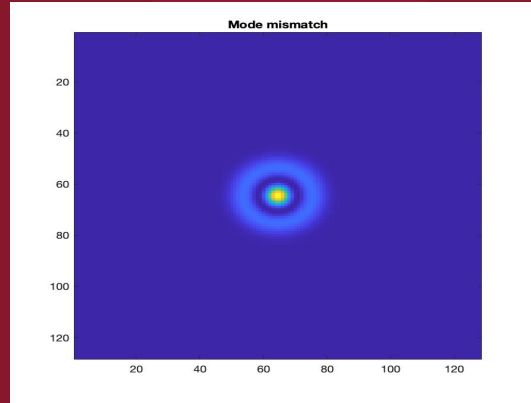
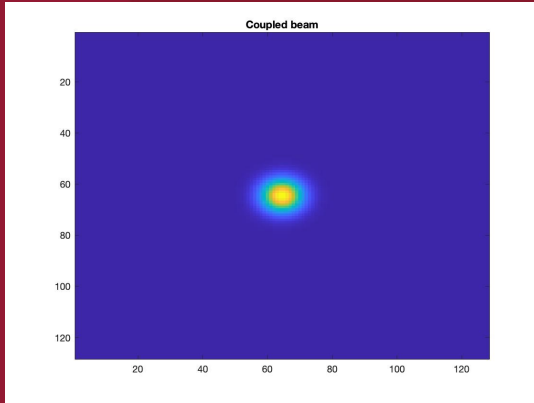


Misalignment

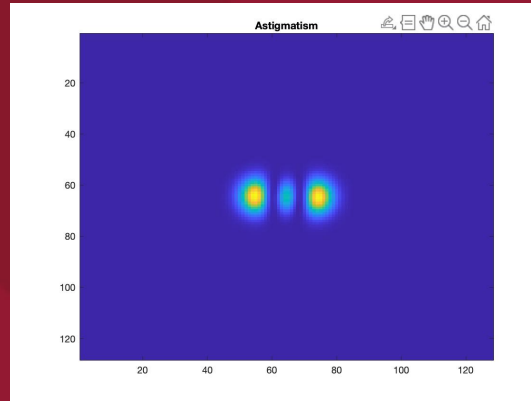
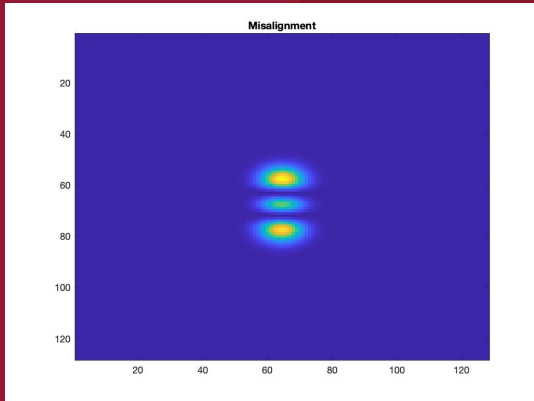
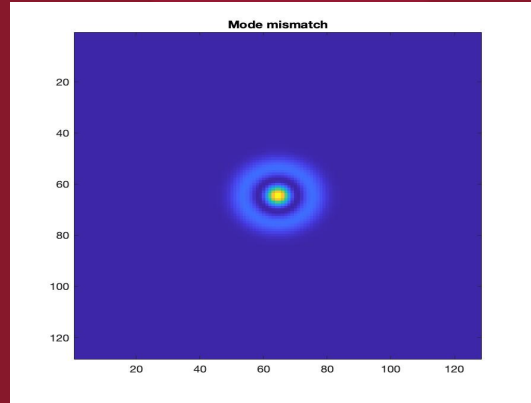
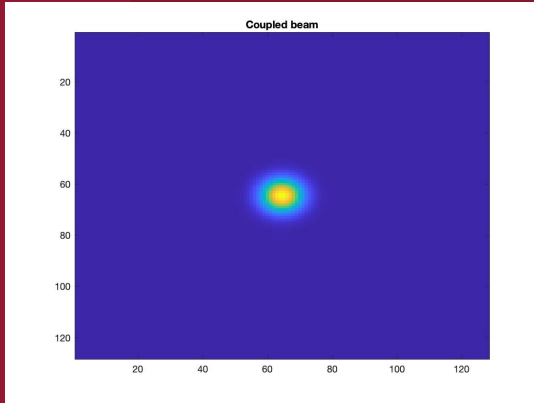


Astigmatism





→ We can easily tell the difference, but not the exact contribution of each parameter



→ We can easily tell the difference, but not the exact contribution of each parameter

→ How then?

2) What is Machine Learning and why do we need it?

Machine learning (ML) is a field devoted to understanding and building methods that let machines "learn".

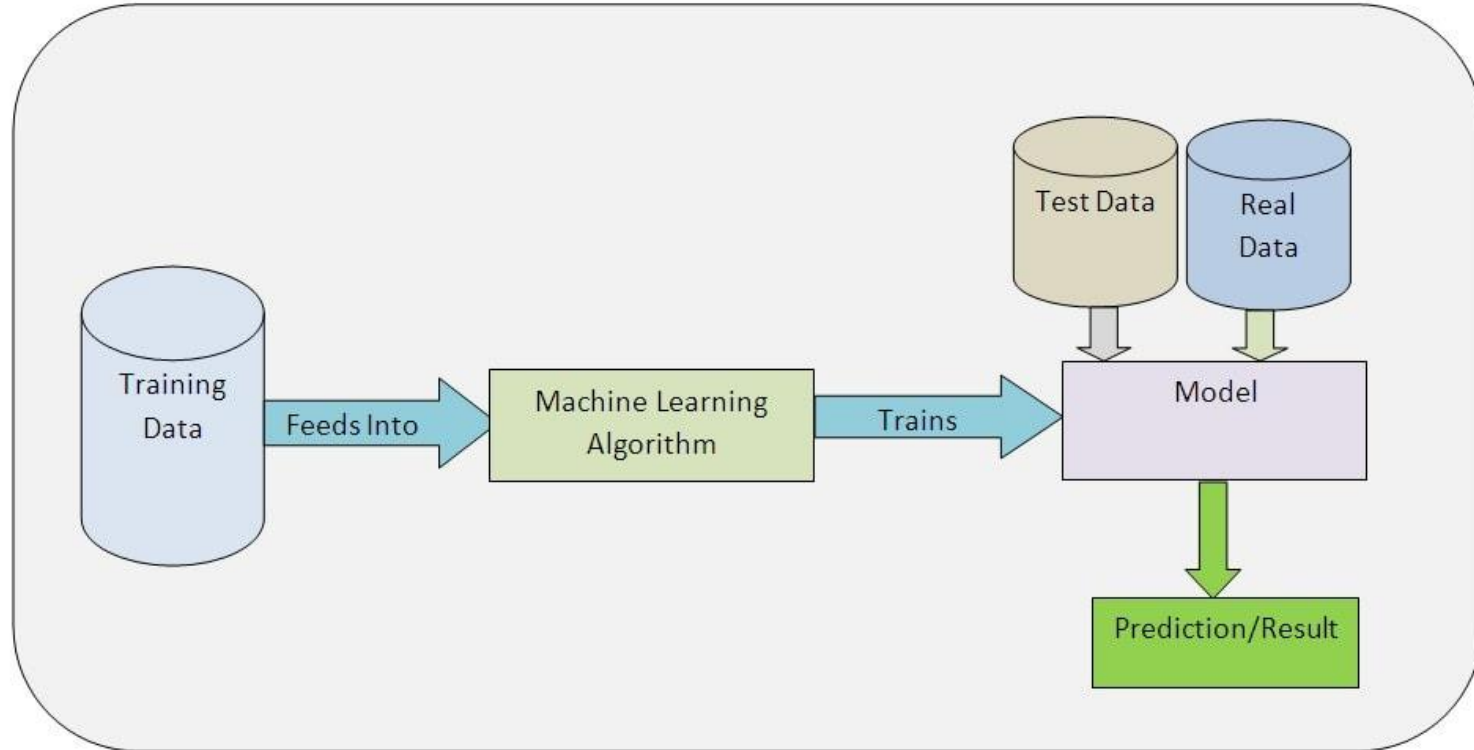
Machine learning (ML) is a field devoted to understanding and building methods that let machines "learn".

Machine learning algorithms build a model based on sample data, known as training data, in order to make predictions or decisions without being explicitly programmed to do so.

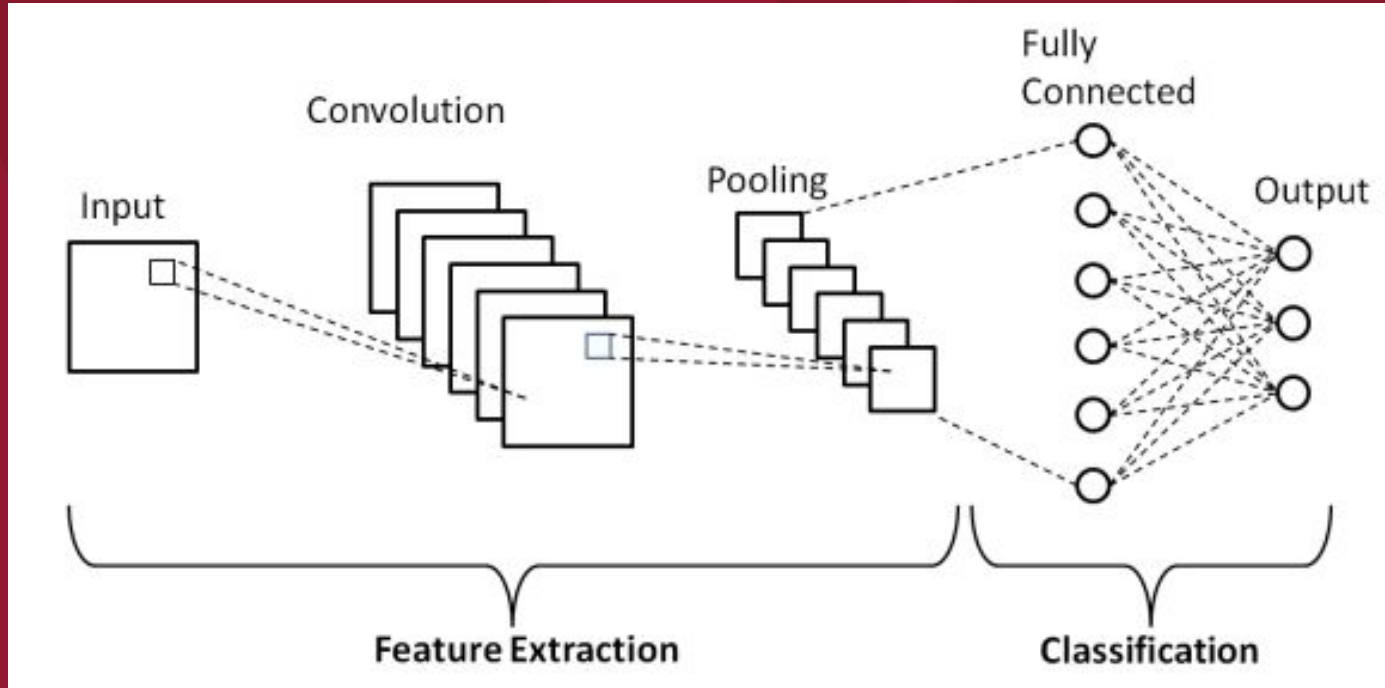
- “Complex fit” (with too many parameters)

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- Useful for evolving systems

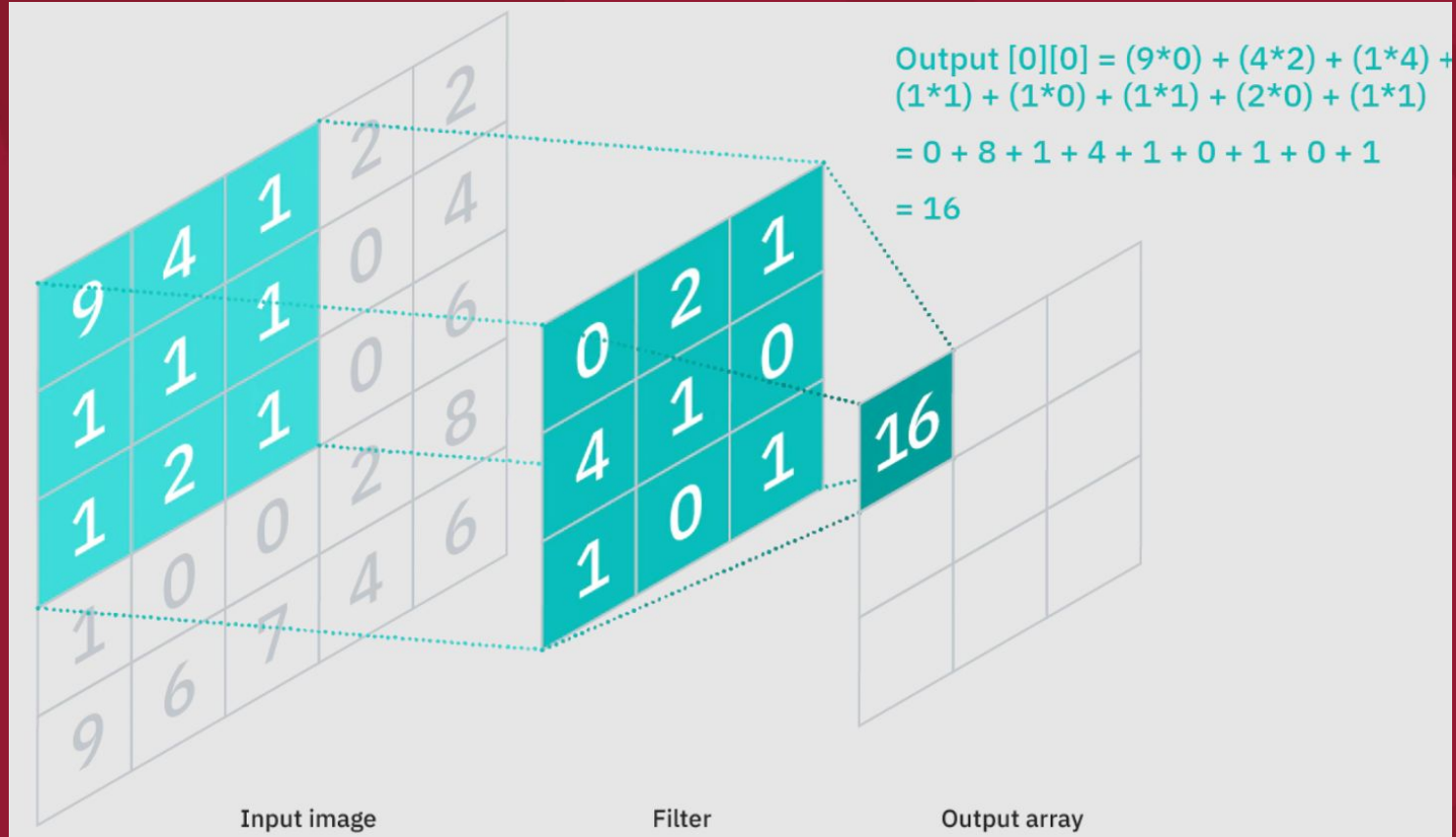
- “Complex fit” (with too many parameters)
- Useful for evolving systems
- Can handle nonlinearity

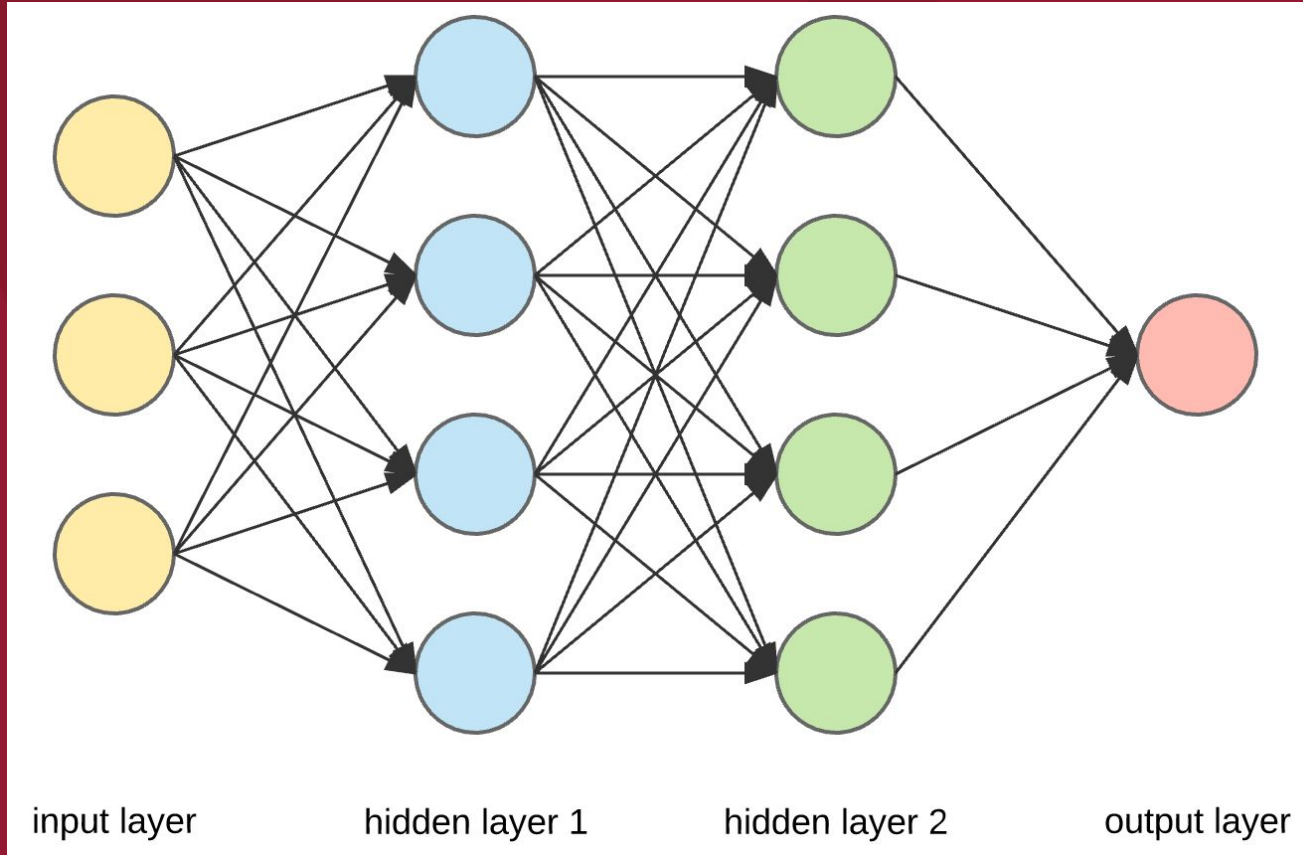


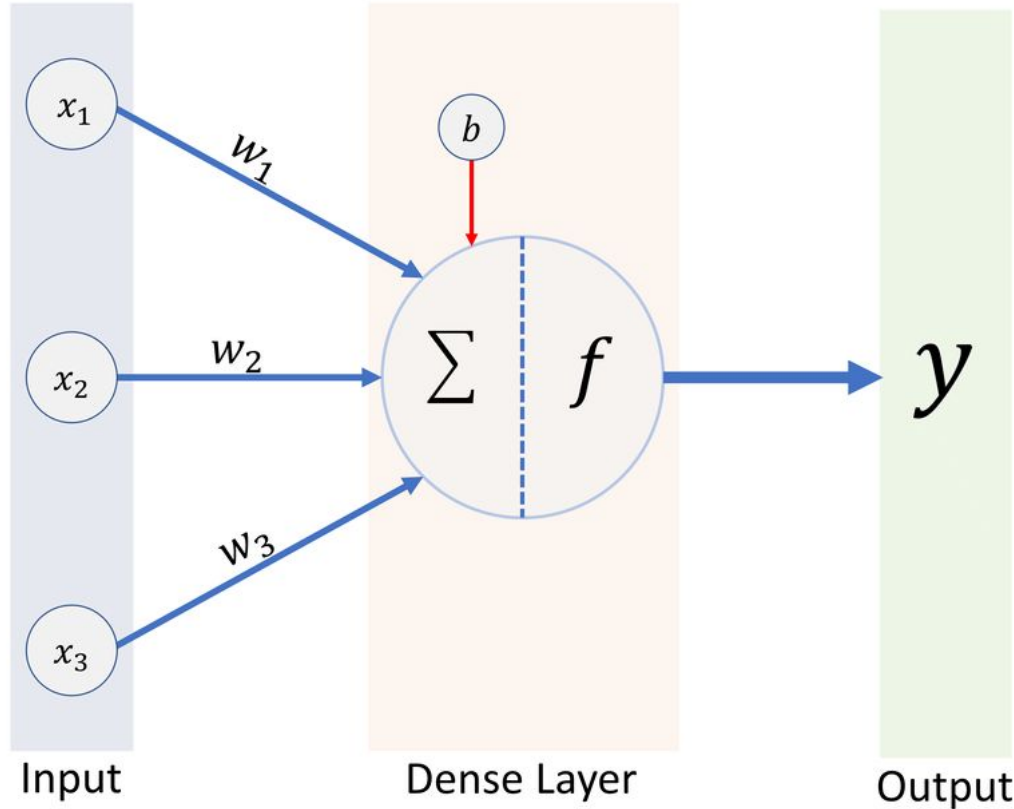
A Simple Machine Learning Pipeline Explanation



What is Machine Learning and why do we need it? 19

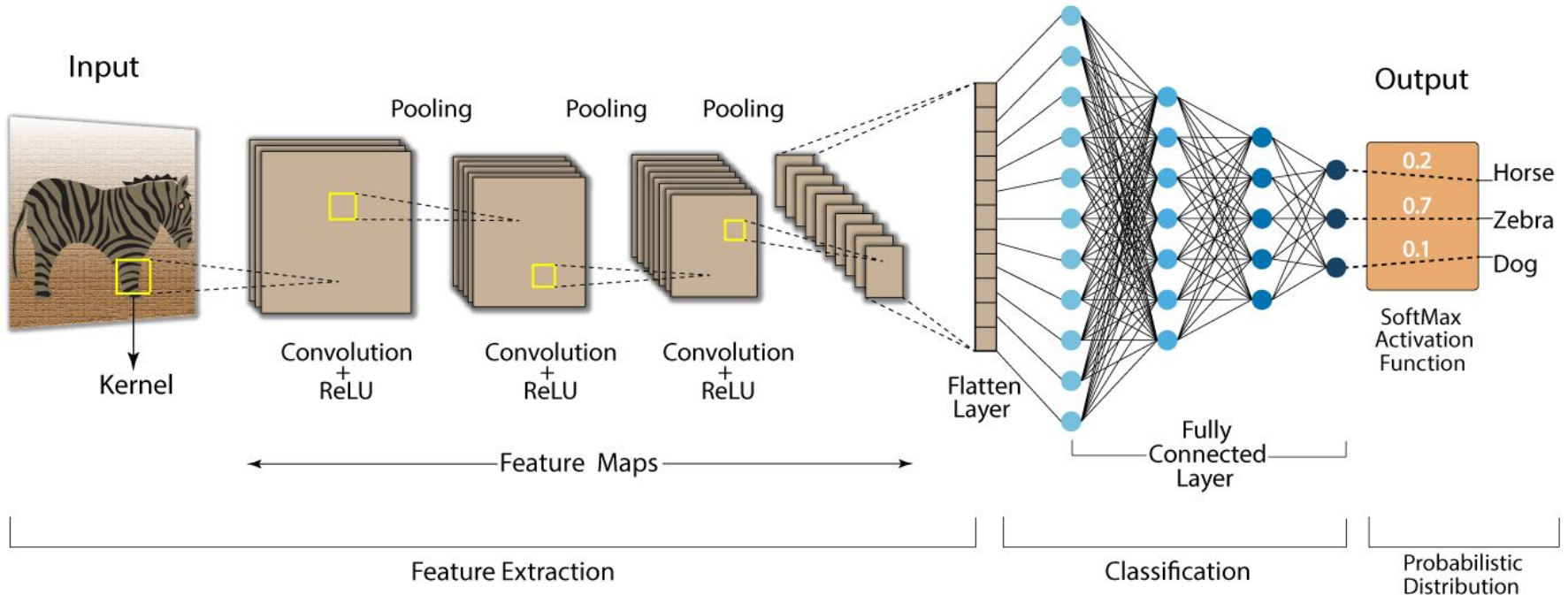






Weighted sum +
activation function

Convolution Neural Network (CNN)



3) The analysis

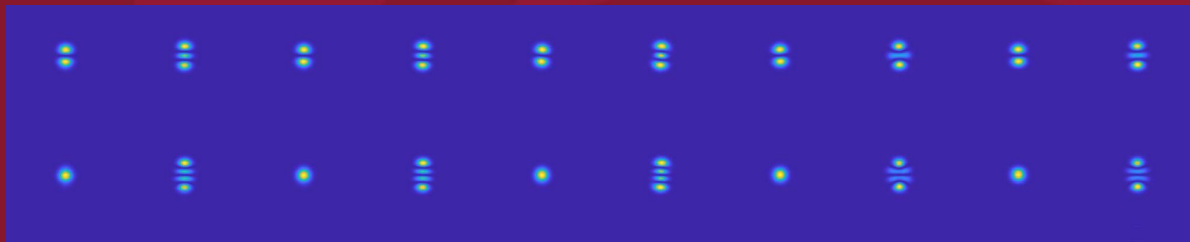
- “Machine Learning for Quantum-Enhanced Gravitational-Wave Observatories” - Chris Whittle, Ge Yang, Matthew Evans, Lisa Barsotti (05/2023)

- “Machine Learning for Quantum-Enhanced Gravitational-Wave Observatories” - Chris Whittle, Ge Yang, Matthew Evans, Lisa Barsotti (05/2023)
- “Predicting the motion of a high-Q pendulum subject to seismic perturbations using machine learning” - Nicolas Heimann, Jan Petermann, Daniel Hartwig, Roman Schnabel, Ludwig Mathey

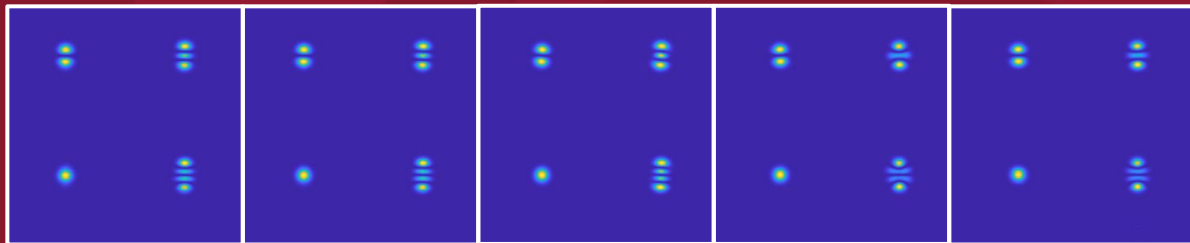
Input:

Input: Cavity scan

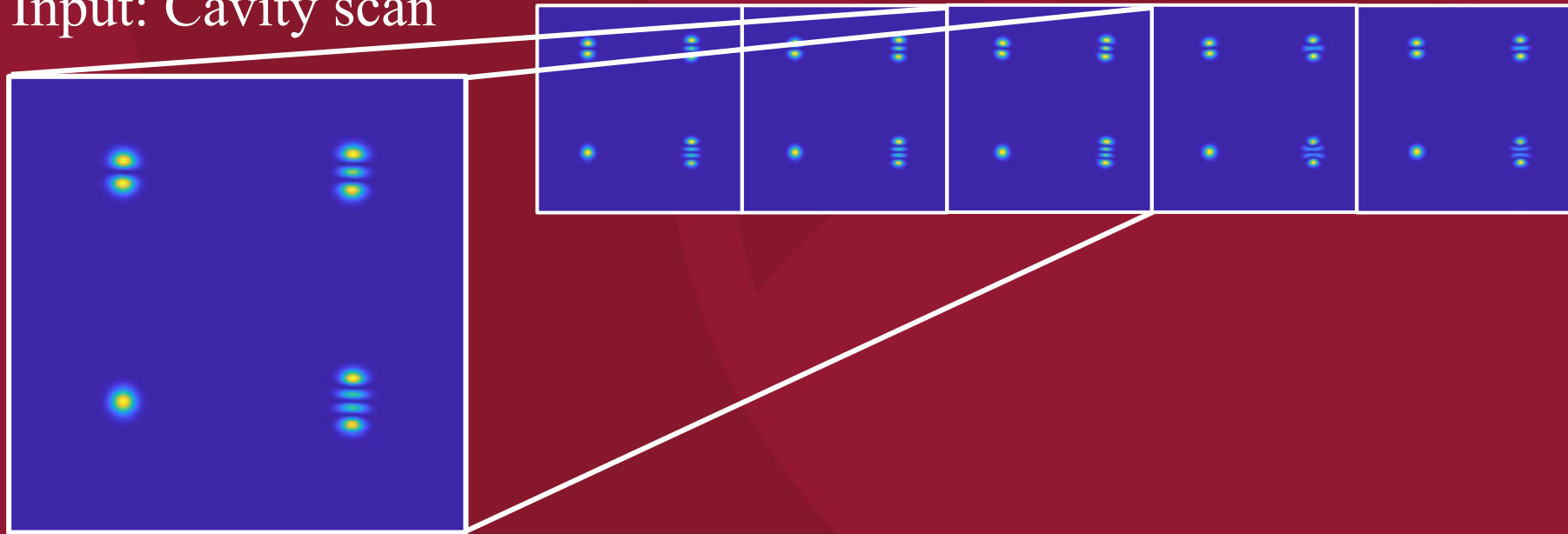
Input: Cavity scan



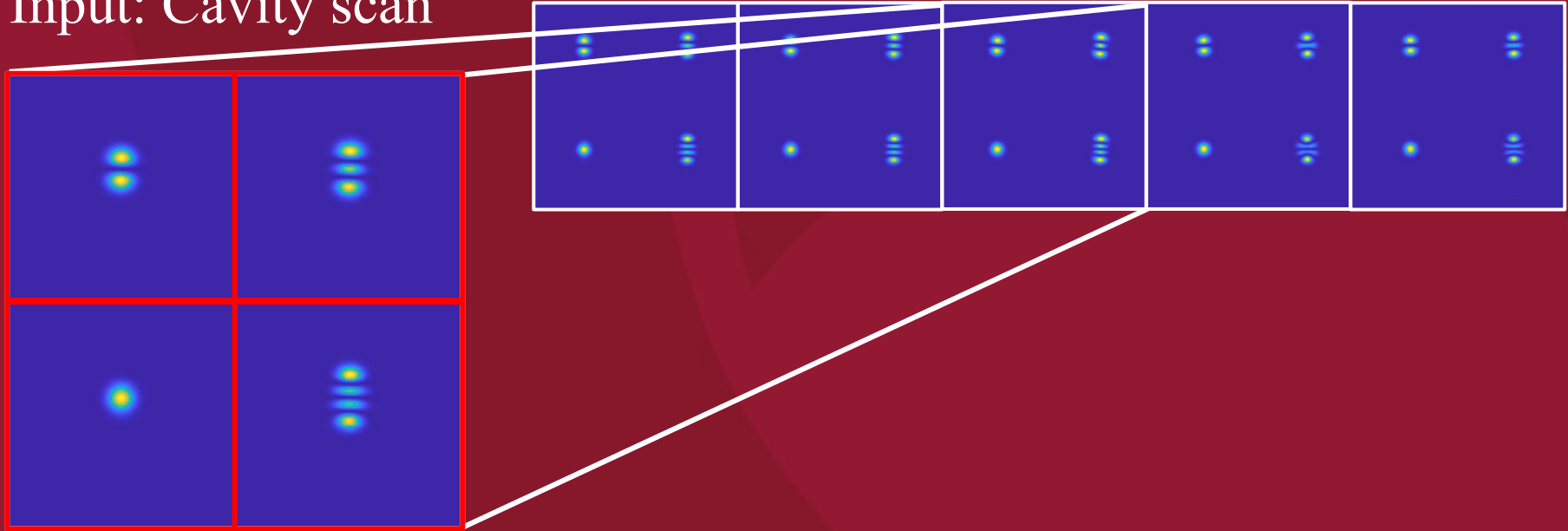
Input: Cavity scan



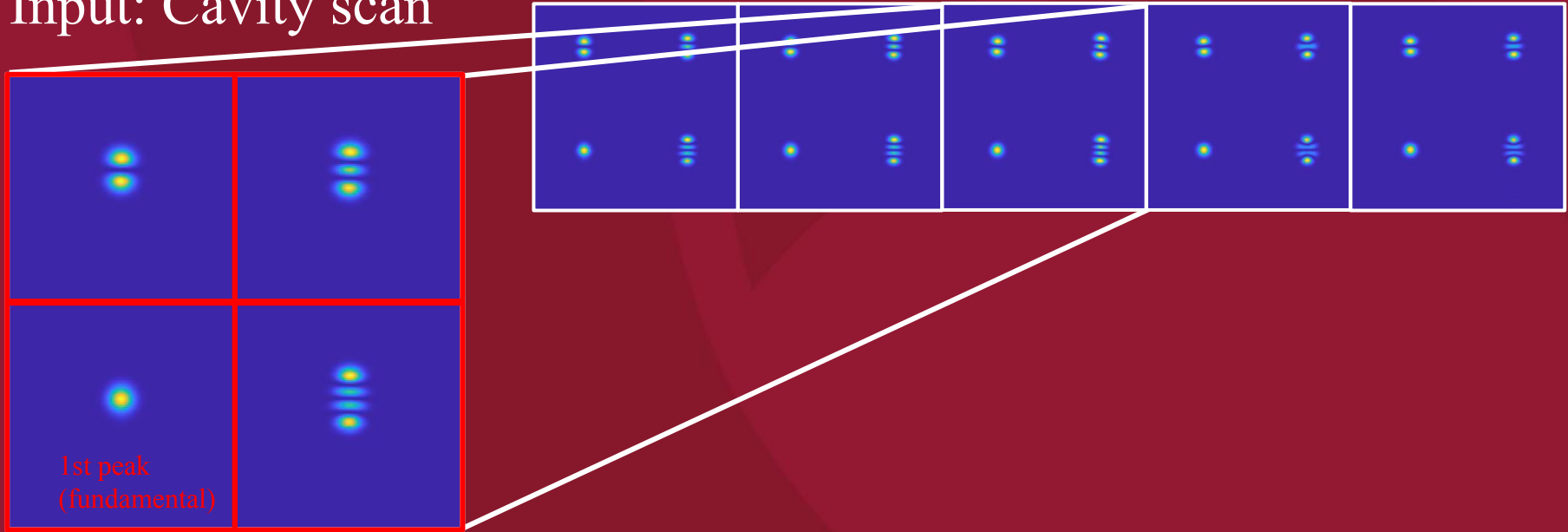
Input: Cavity scan



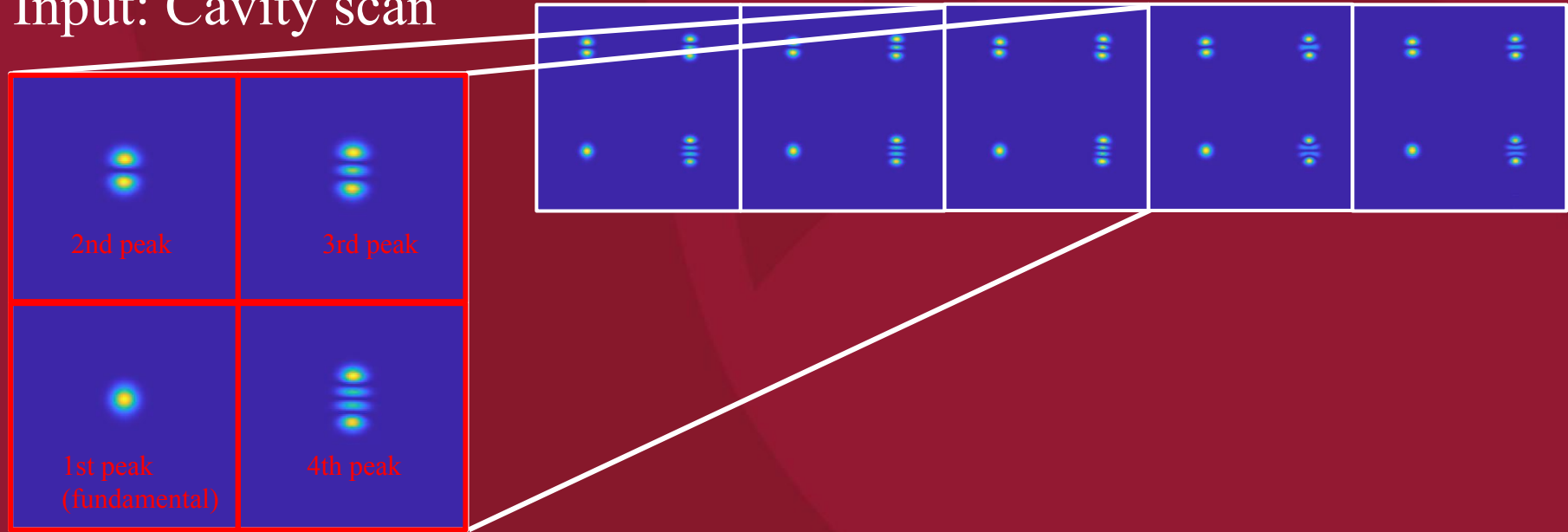
Input: Cavity scan



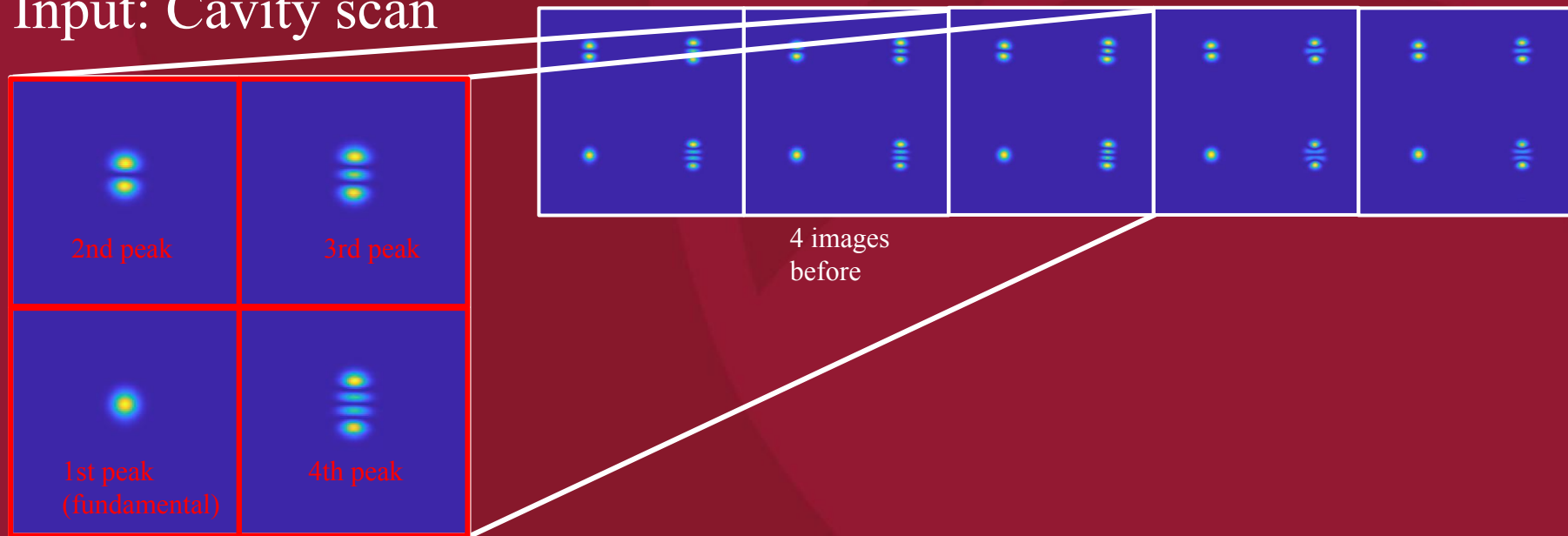
Input: Cavity scan



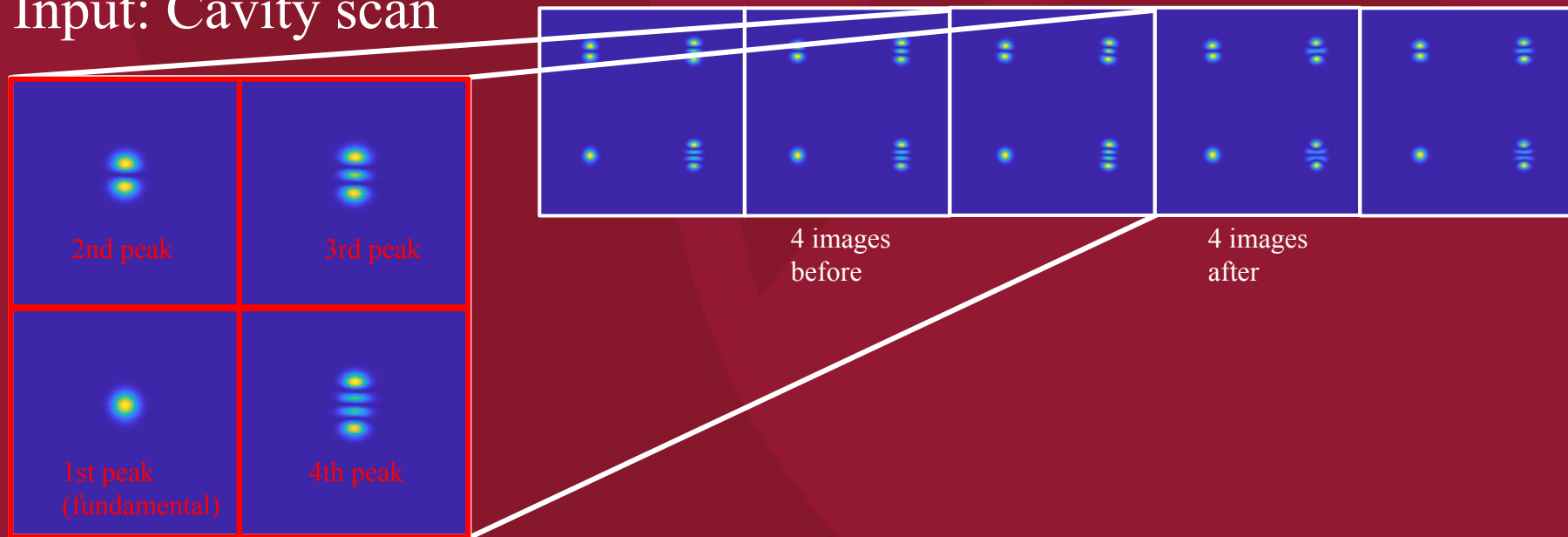
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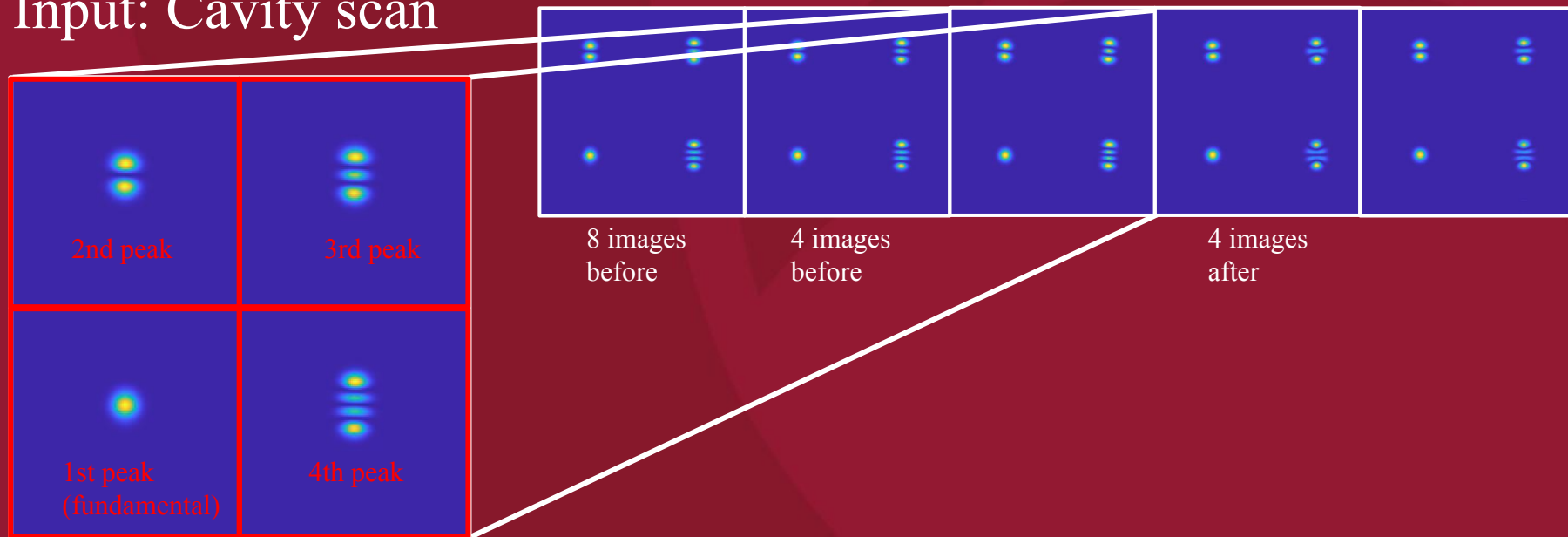
Input: Cavity scan



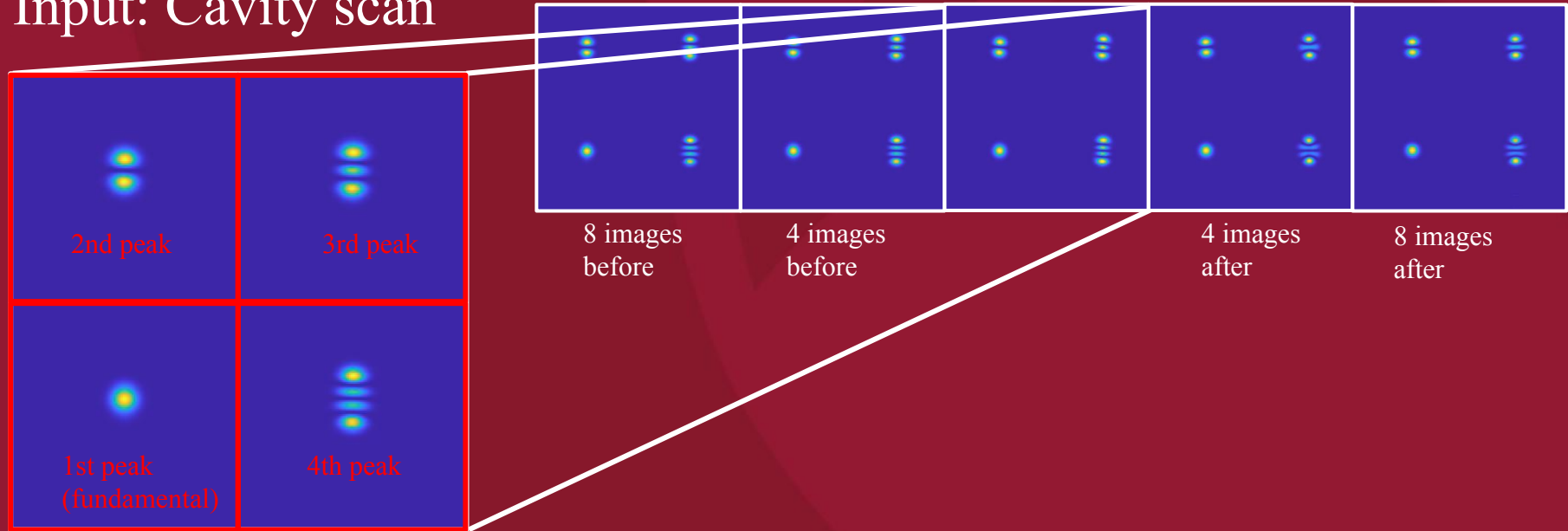
Input: Cavity scan



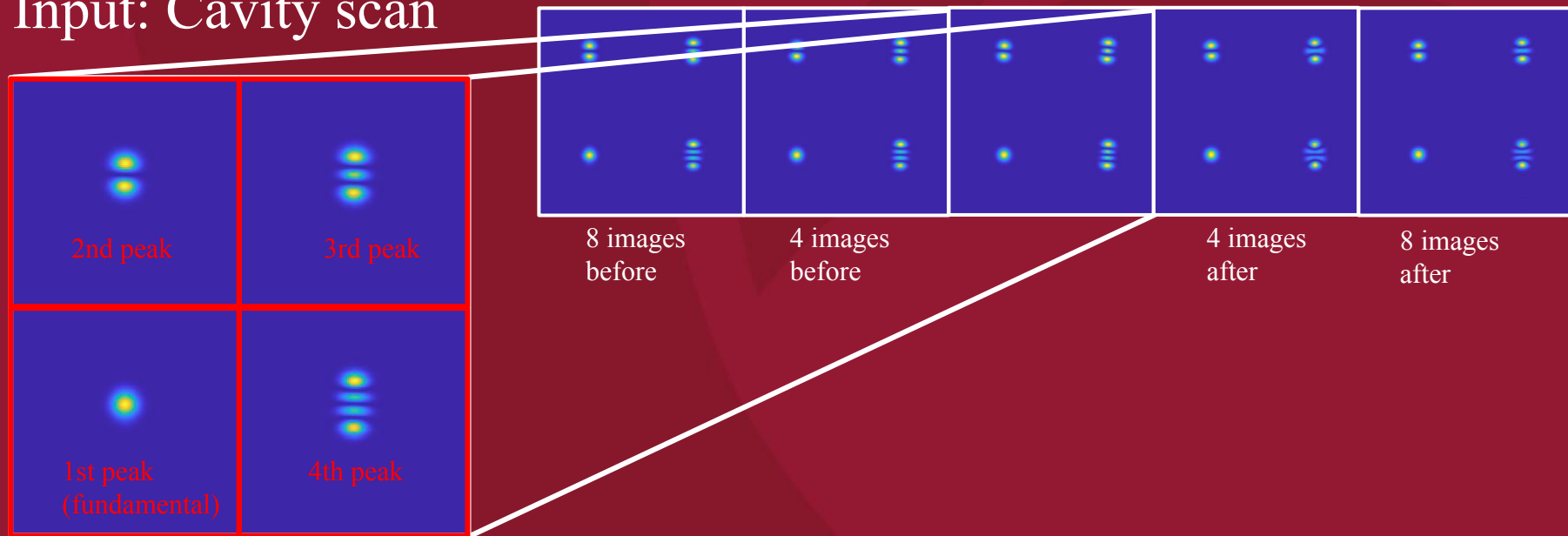
Input: Cavity scan



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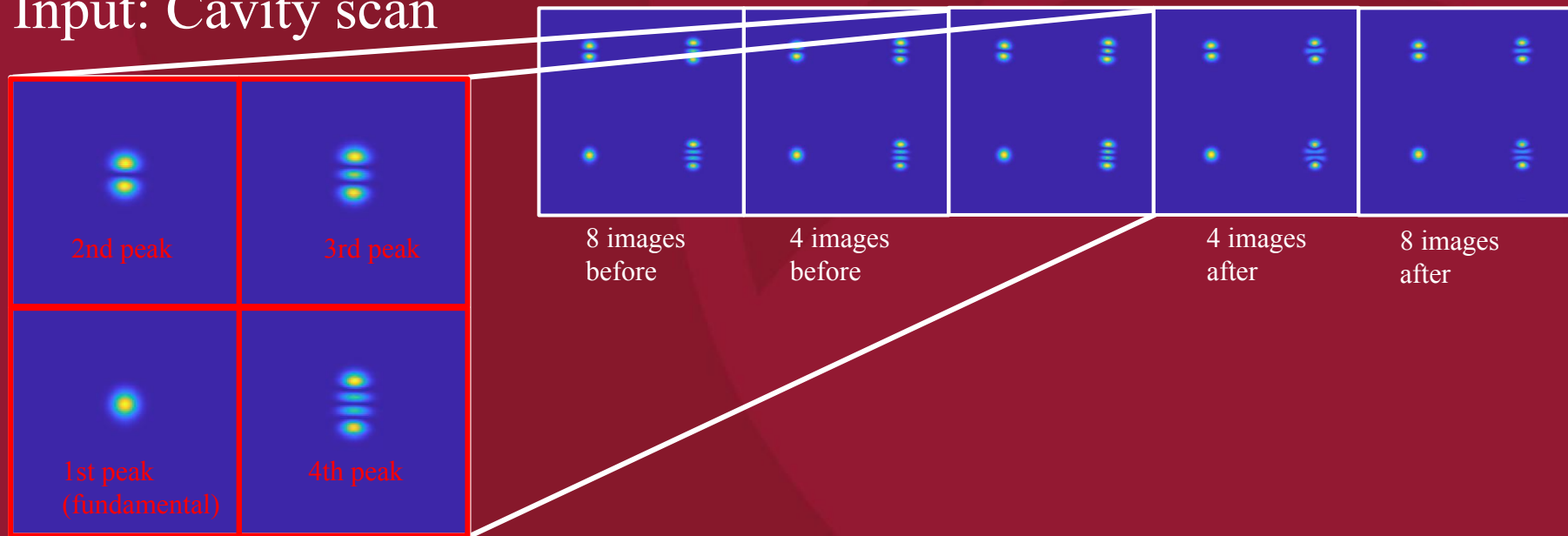


Input: Cavity scan



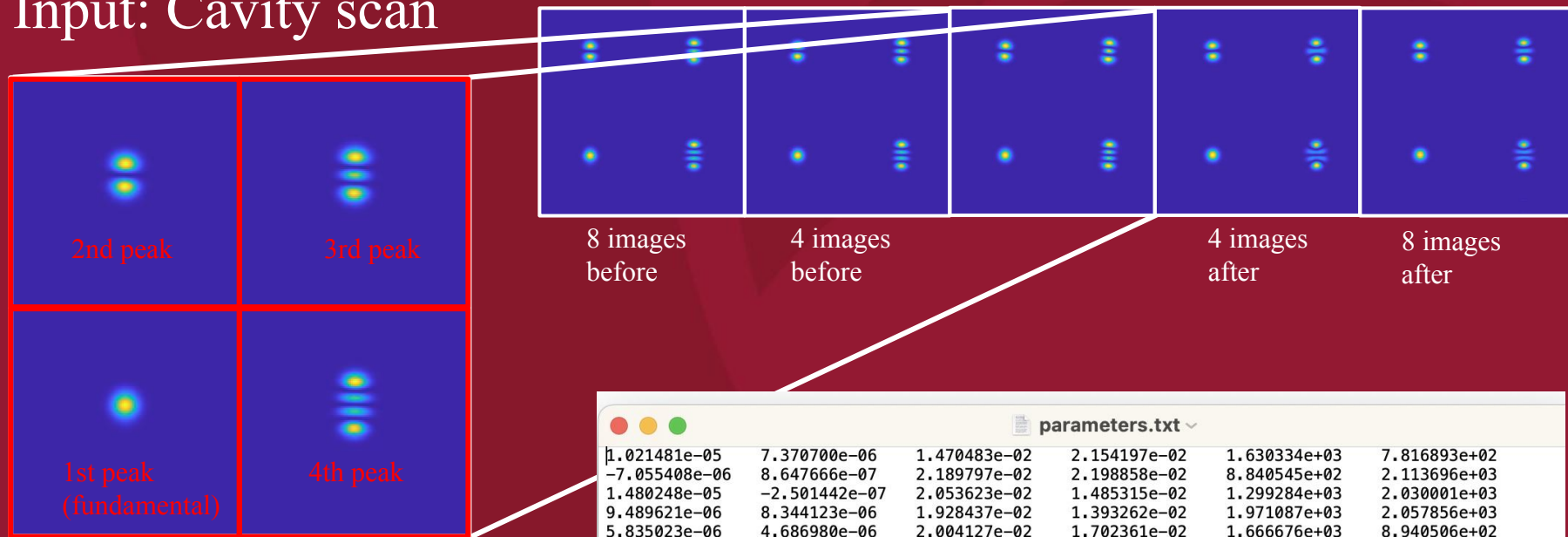
Output:

Input: Cavity scan



Output: 6 parameters

Input: Cavity scan

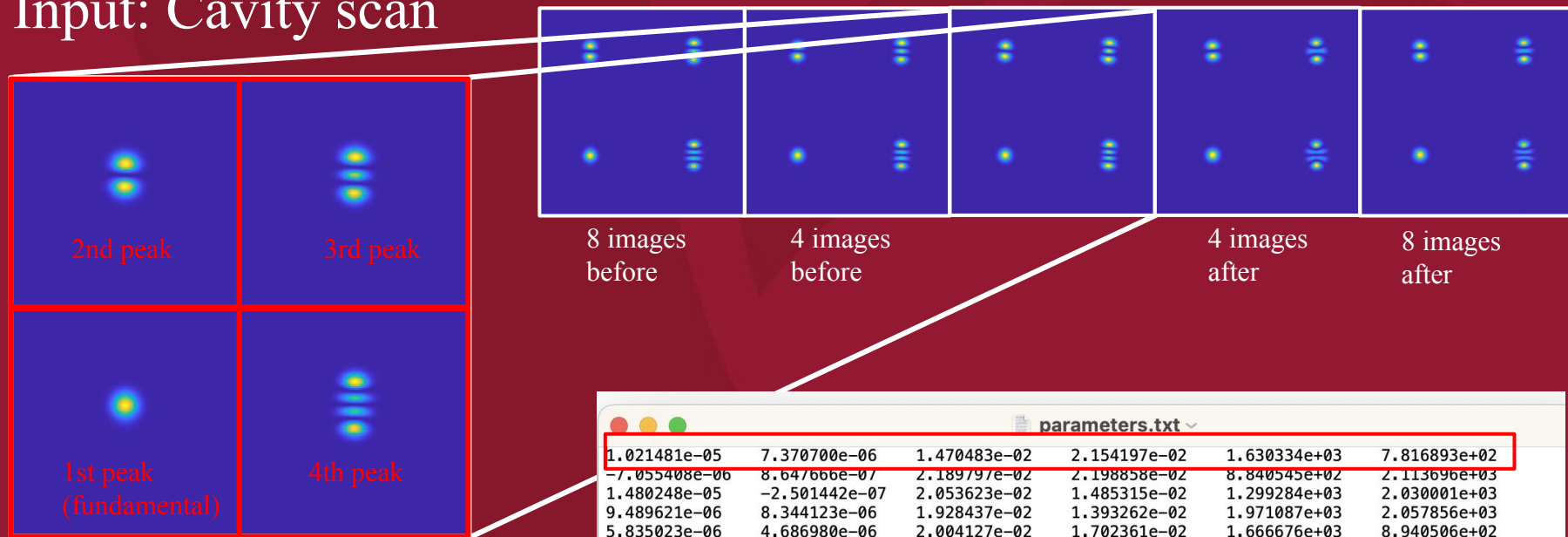


Output: 6 parameters

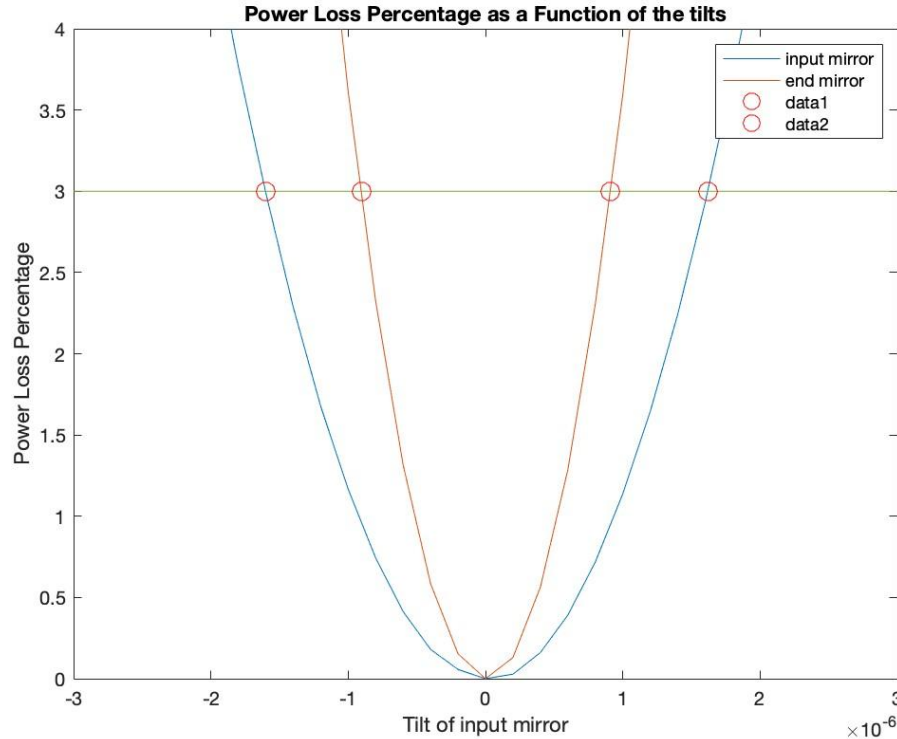
parameters.txt

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1.480248e-05	-2.501442e-07	2.053623e-02	1.485315e-02	1.299284e+03	2.030001e+03
9.489621e-06	8.344123e-06	1.928437e-02	1.393262e-02	1.971087e+03	2.057856e+03
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6.714626e-06	-8.471469e-06	1.600343e-02	1.402331e-02	7.889807e+02	1.889214e+03
6.353345e-06	-3.300469e-06	2.183487e-02	1.392165e-02	1.325980e+03	1.215013e+03
8.629999e-06	5.366009e-06	1.522350e-02	1.786926e-02	1.336735e+03	1.618946e+03
6.821513e-06	4.631630e-06	1.599565e-02	1.951602e-02	1.666078e+03	8.809679e+02
-1.219244e-05	-1.470467e-08	2.191734e-02	1.657414e-02	1.556308e+03	9.743404e+02
8.171058e-06	-4.424416e-06	1.798709e-02	1.968400e-02	2.036754e+03	2.096454e+03
1.599171e-06	-6.535669e-06	1.489804e-02	1.585560e-02	1.957863e+03	1.020829e+03
1.020067e-05	-4.634147e-06	2.165335e-02	1.665736e-02	9.453327e+02	1.015949e+03
3.815951e-06	-4.692404e-07	1.665072e-02	2.082628e-02	1.556303e+03	1.471580e+03

Input: Cavity scan

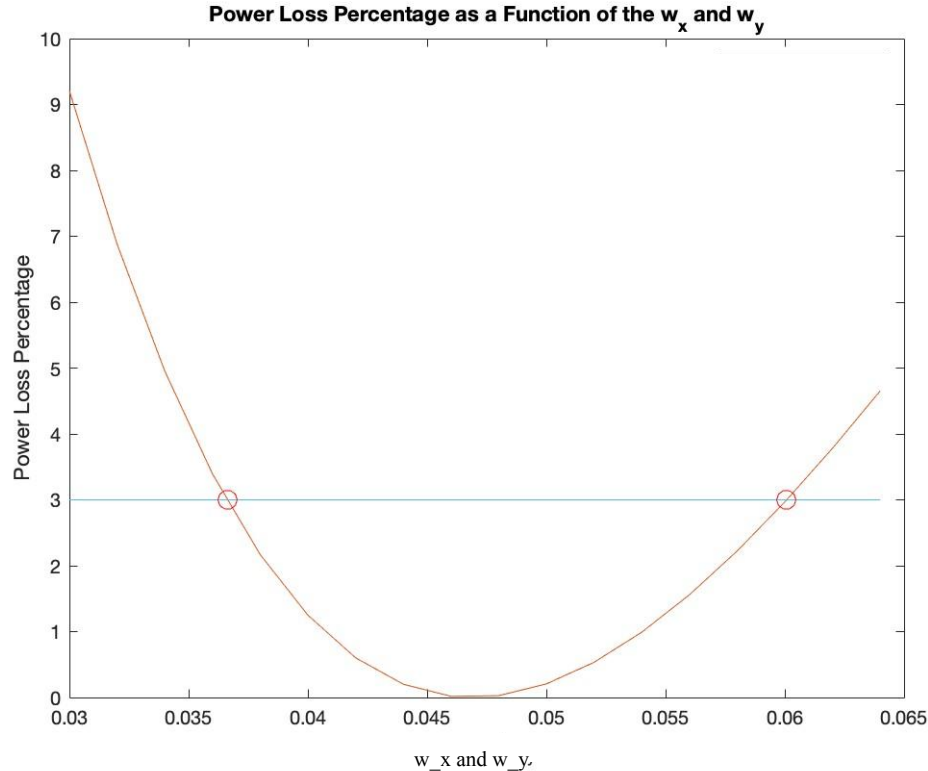


Output: 6 parameters



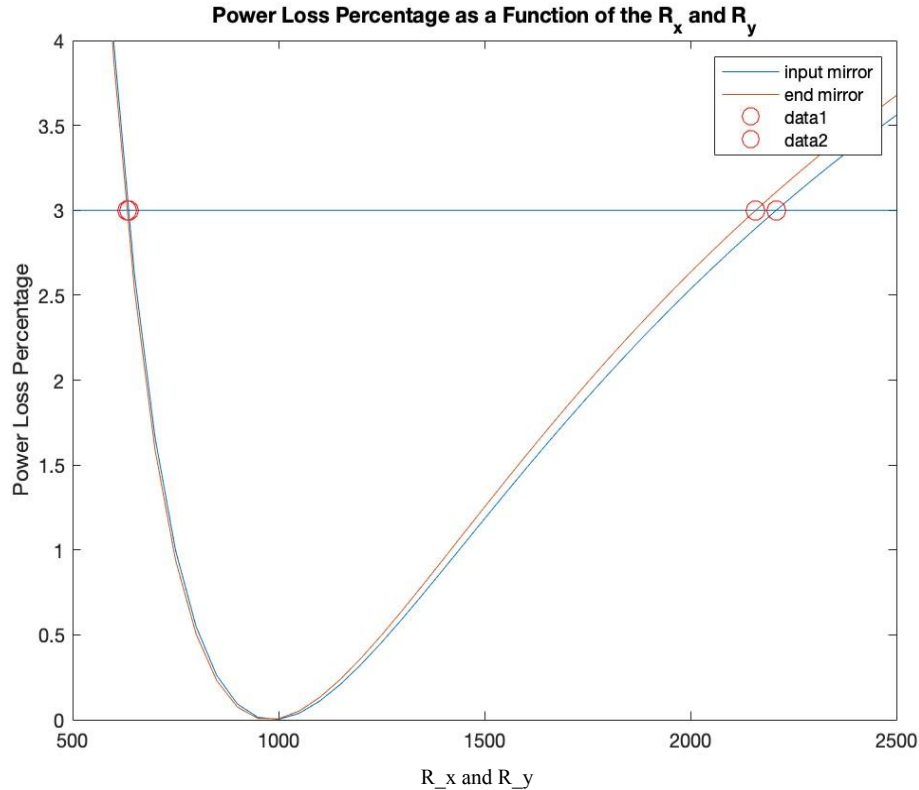
Intersection points for input mirror:
(-1.6025e-06, 3)
(1.6182e-06, 3)

Intersection points for end mirror:
(-9.0485e-07, 3)
(9.0784e-07, 3)



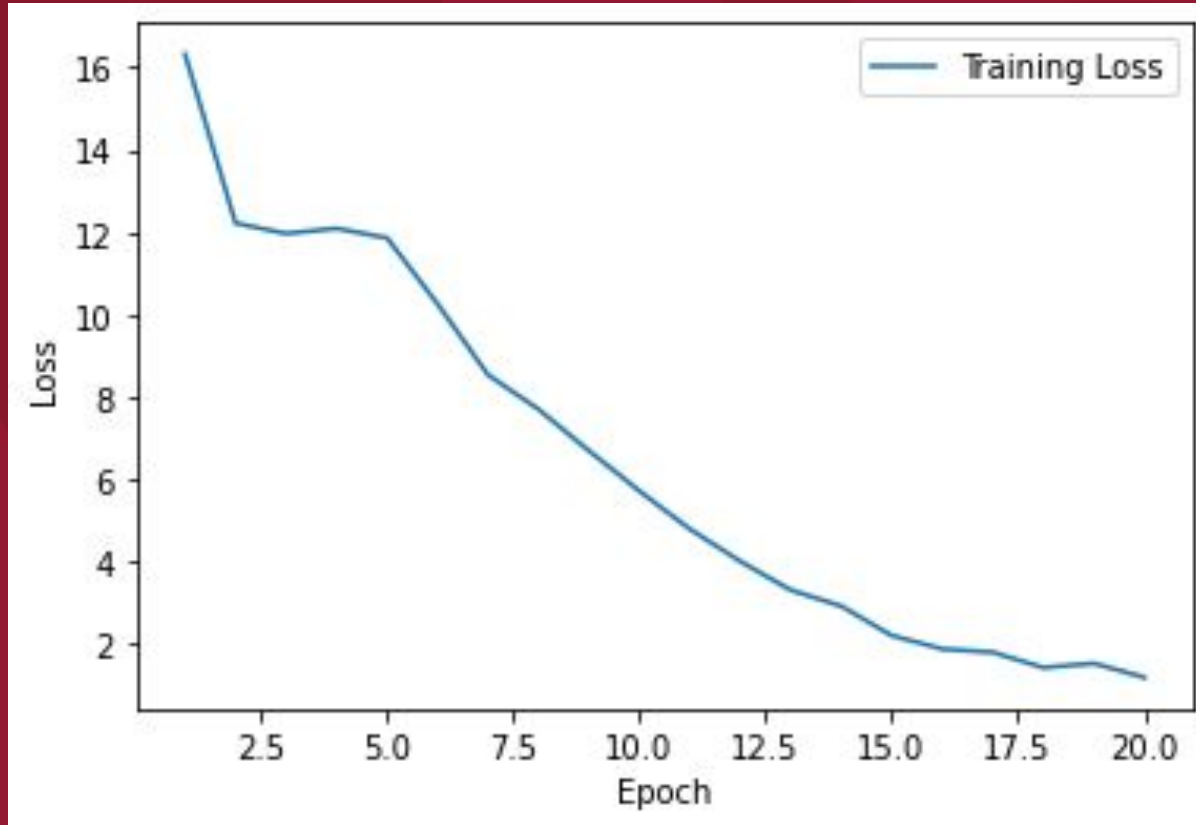
Intersection points for w_x :
(0.036643, 3)
(0.060077, 3)

Intersection points for w_y :
(0.036643, 3)
(0.060077, 3)



Intersection points for R_x :
(636.2939, 3)
(2208.2487, 3)

Intersection points for R_y :
(632.8729, 3)
(2158.5623, 3)



Variable number of inputs
(number of peaks)

Variable number of inputs
(number of peaks)



Truncation

Variable number of inputs
(number of peaks)



Truncation



Loss of data

Variable number of inputs
(number of peaks)



Truncation



Padding

Loss of data

Variable number of inputs
(number of peaks)



Truncation



Loss of data

Padding

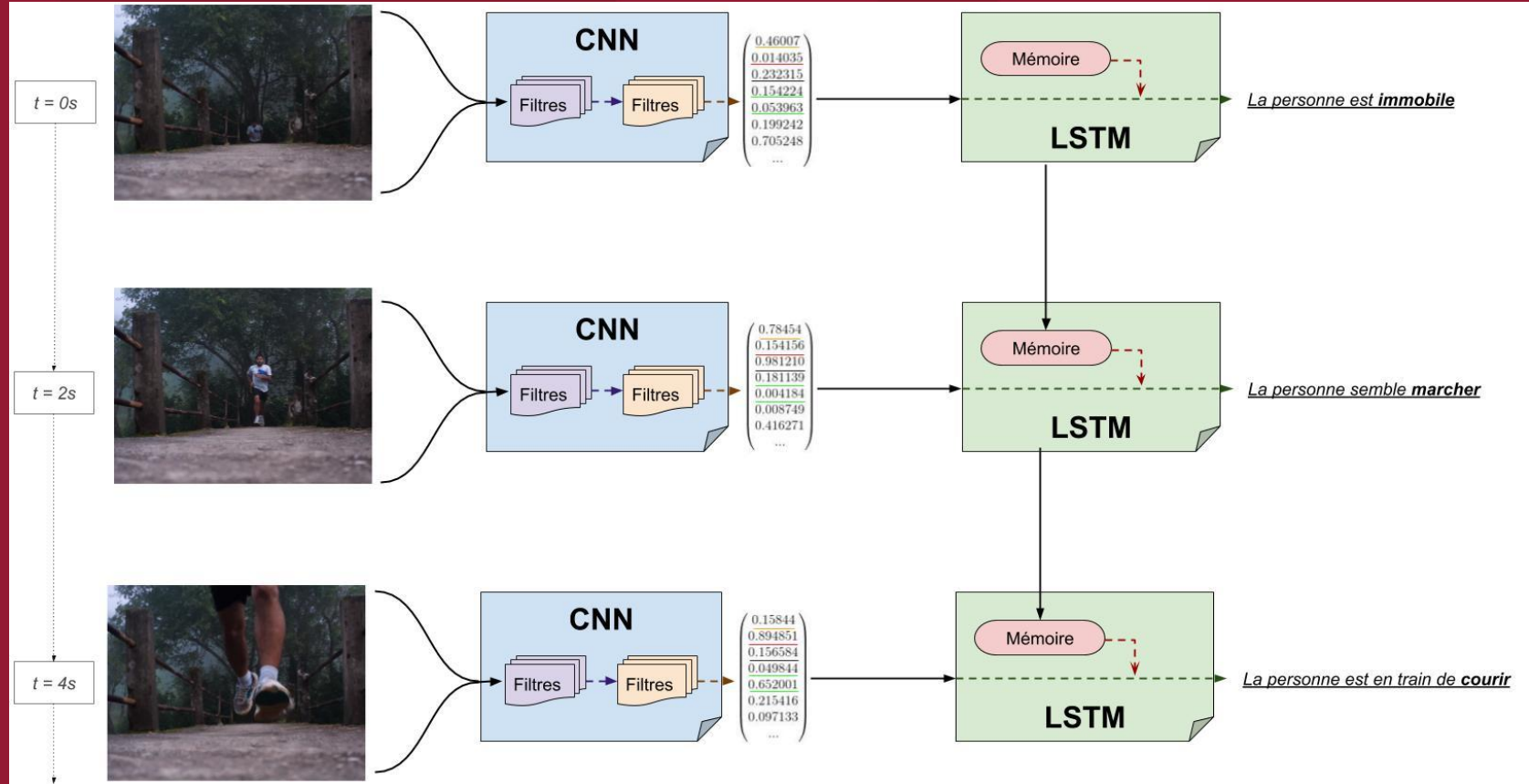


Extra worthless data

Long Short-Term Memory Convolutional Neural Network (LSTM CNN)

Long Short-Term Memory Convolutional Neural Network (LSTM CNN)





Thank you for listening!

